

Topological Data Analysis

Lecture 12

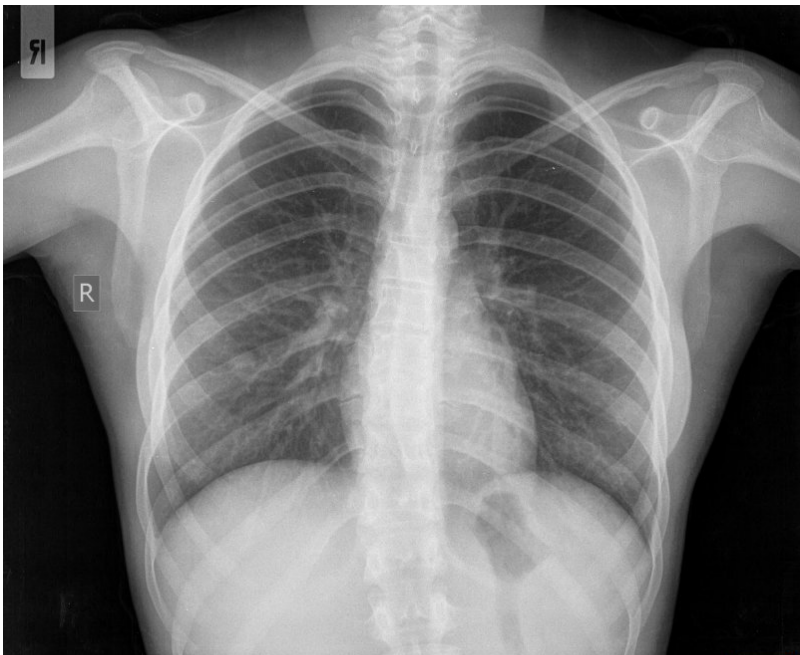
Topological Data Analysis of Digital Images

Oleg Kachan

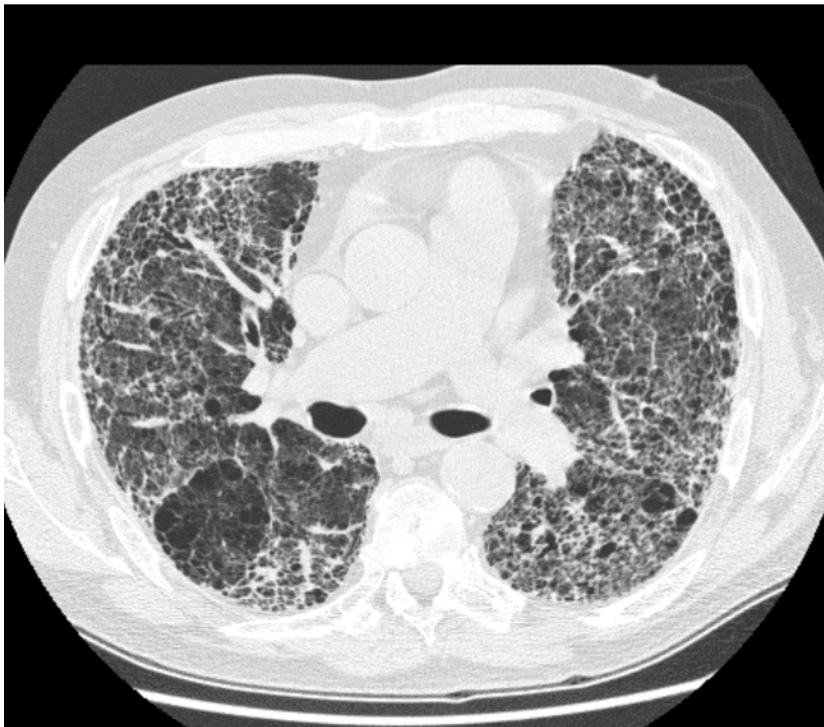
Digital images

Medical imaging

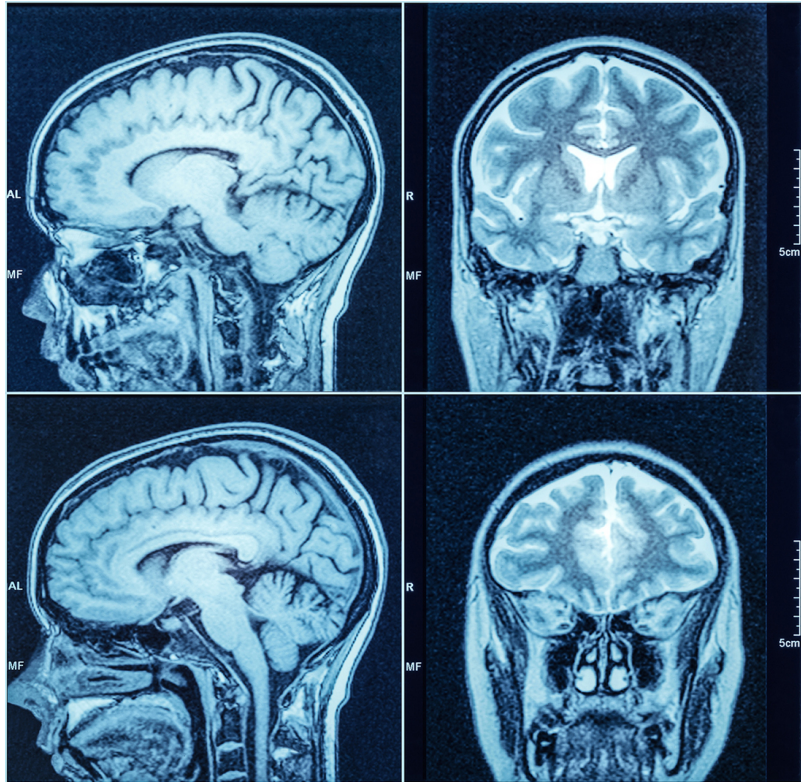
RX



CT

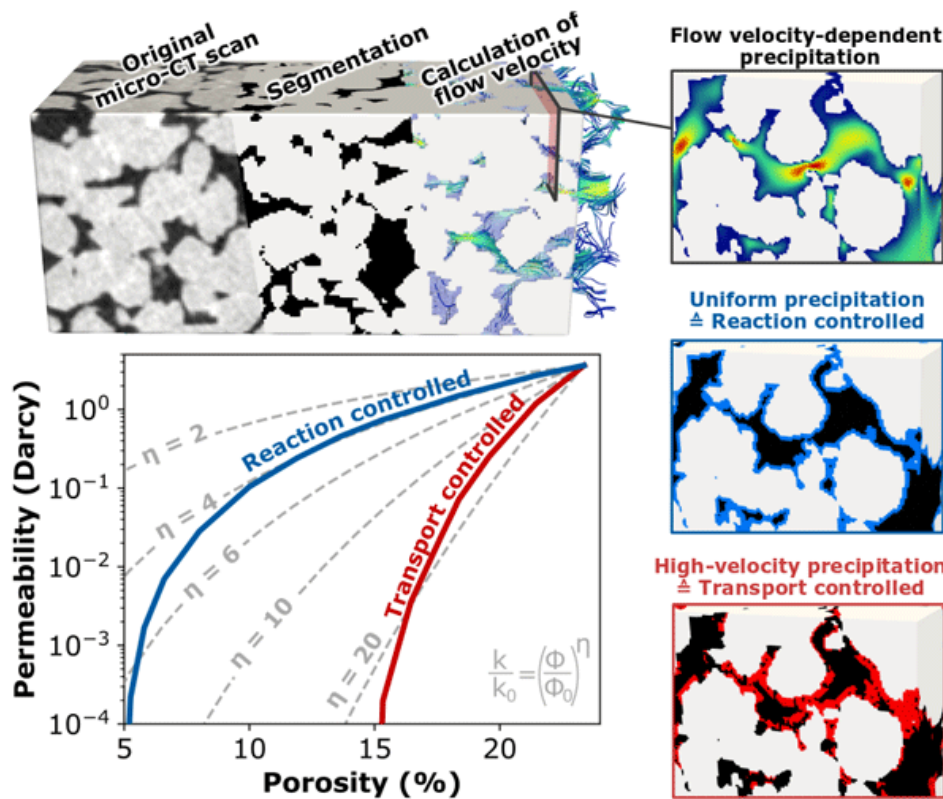


MRI

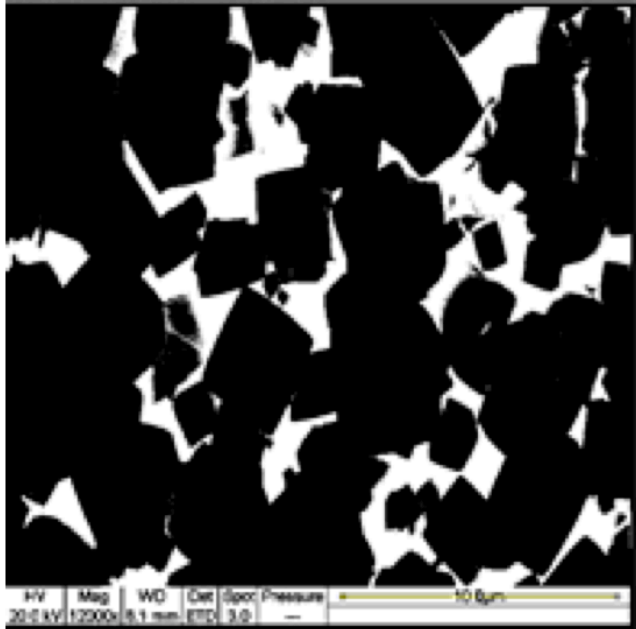
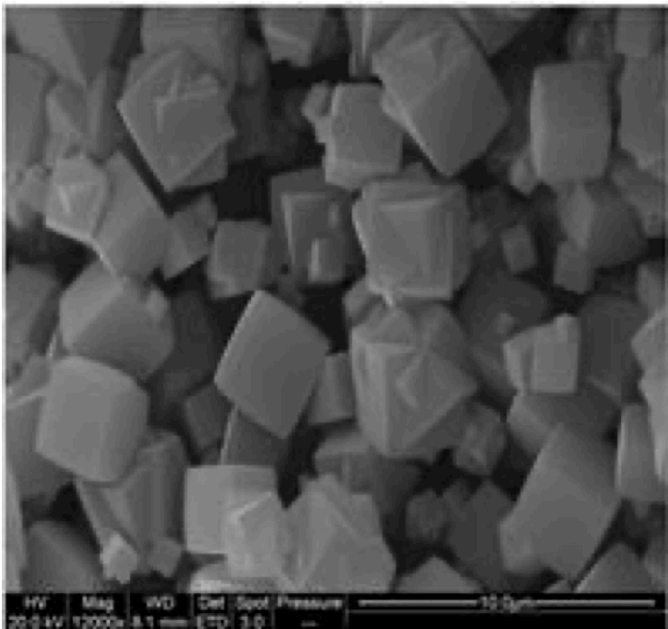


Material science

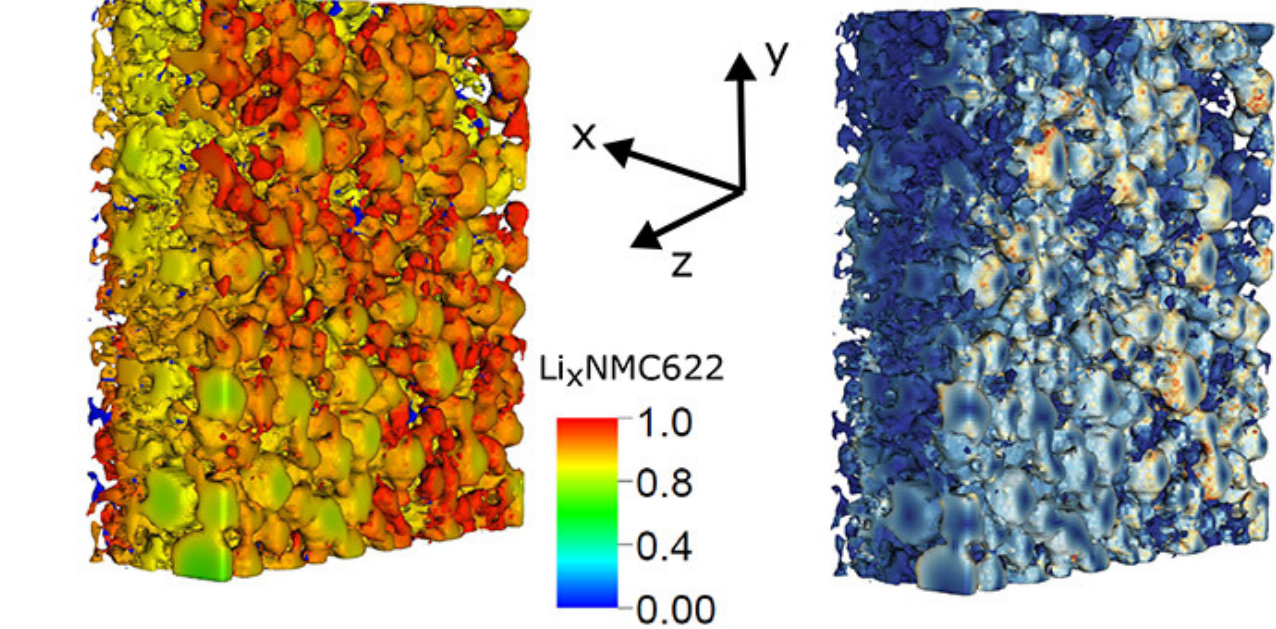
Porous media



Catalysts

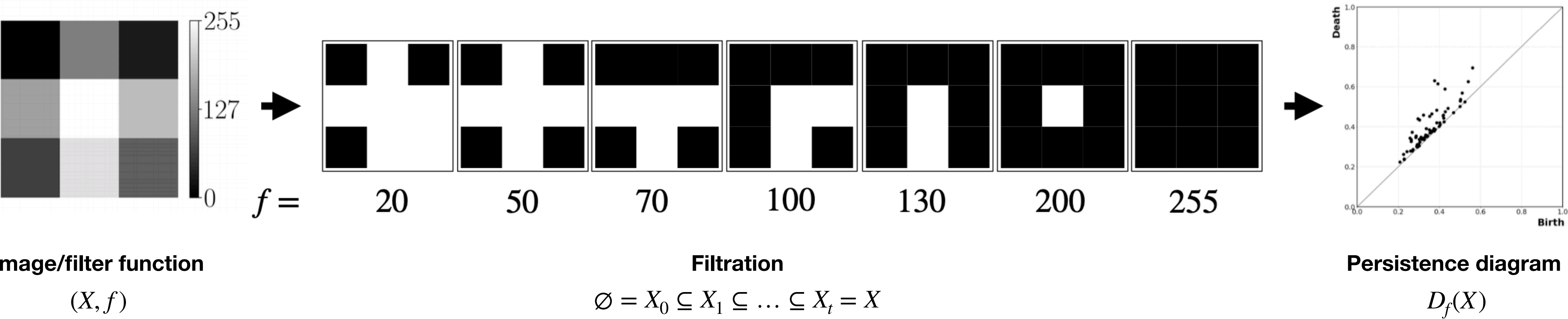


Batteries



Digital images

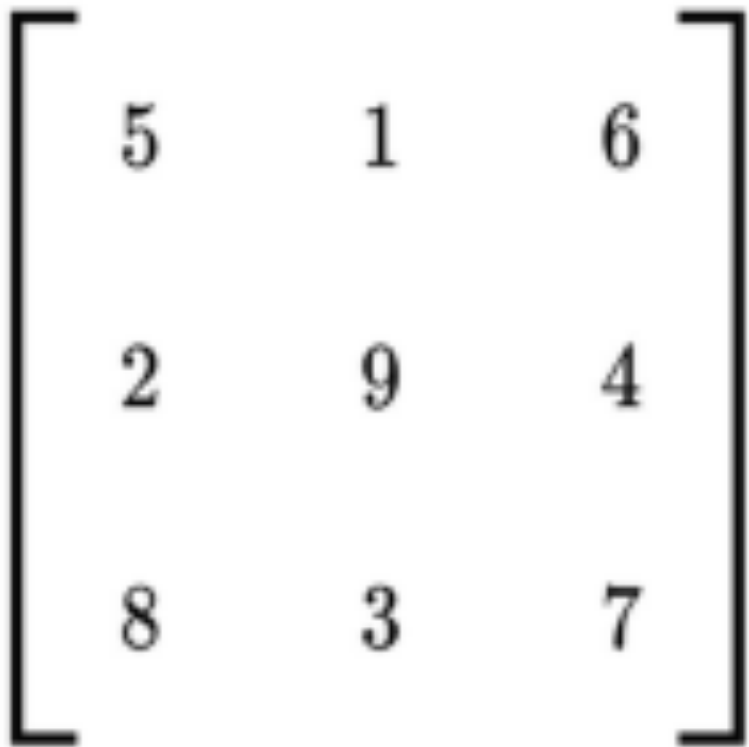
Topological Data Analysis



Digital images

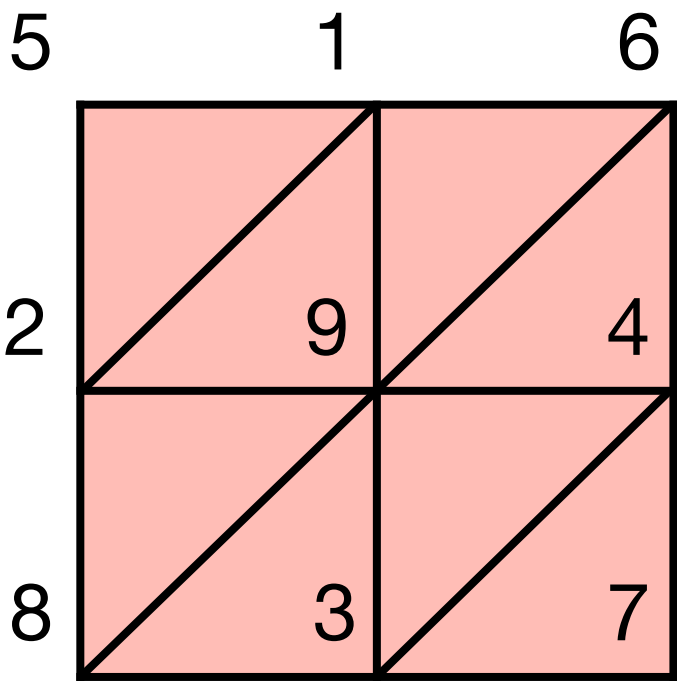
Triangulation

A d-dimensional image is a map $\mathcal{I} : I \subseteq \mathbb{Z}^d \rightarrow \mathbb{R}_{\geq 0}$
 $\mathcal{I}(x)$ is an intensity of a voxel x .



Image

$$\mathcal{I} : I \rightarrow \mathbb{R}$$



Simplicial complex K

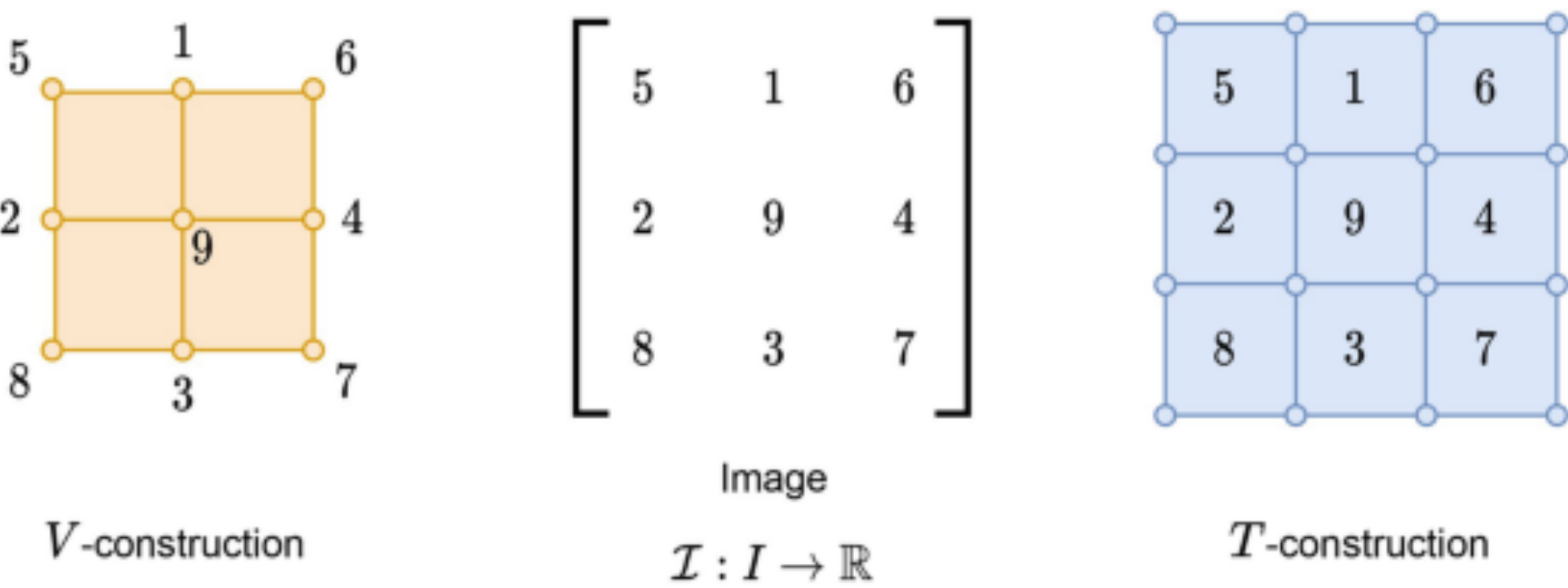
$$f_I : K \rightarrow \mathbb{R}_{\geq 0}$$

$$f(\sigma) = \begin{cases} I, & \dim(\sigma) = 0, \\ \max_{x \sim \sigma} I(v), & \dim \sigma \geq 1. \end{cases}$$

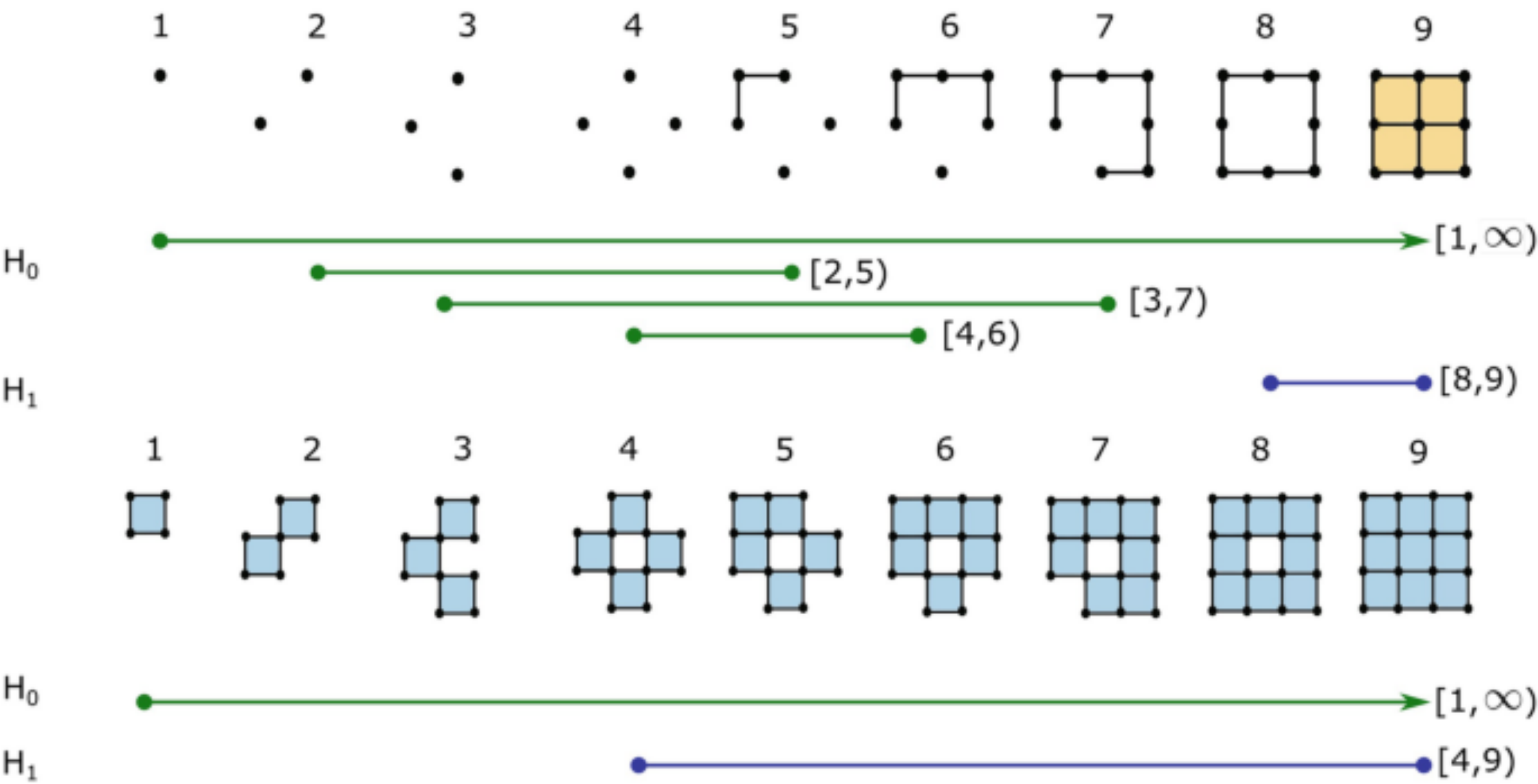
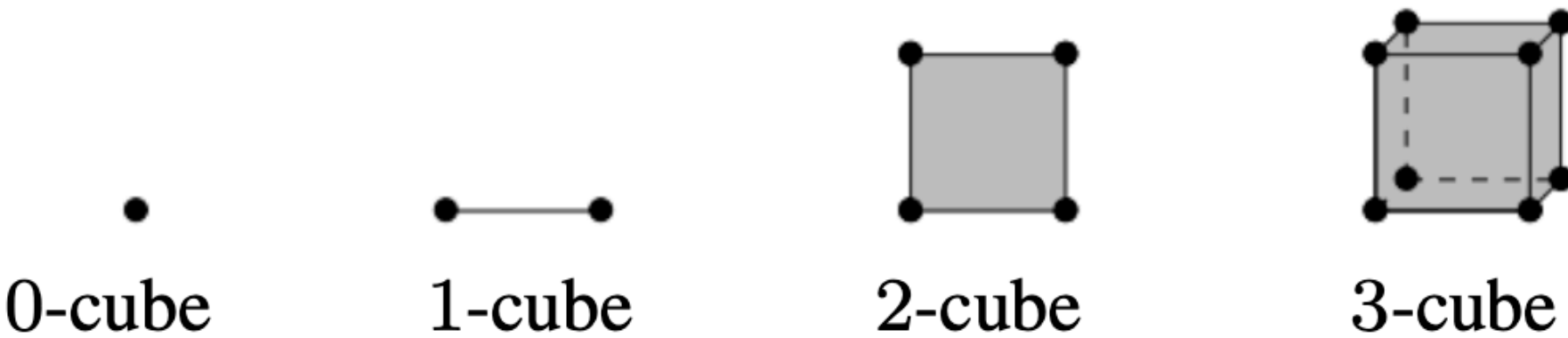
Digital images

Cubical complex

A d-dimensional image is a map $\mathcal{I} : I \subseteq \mathbb{Z}^d \rightarrow \mathbb{R}_{\geq 0}$
 $\mathcal{I}(x)$ is an intensity of a voxel x .



Elementary cubes



Persistent homology of images

Filtration functions

Intensity

$f(x)$ is given by the intensity $\mathcal{I}(x)$ of the image.

Direction

$$f_v(x) = d(x, H(v)),$$

where v is a direction vector.

$H(v)$ is a hyperplane defined by v .

Radial

$$f_c(x) = d(x, c),$$

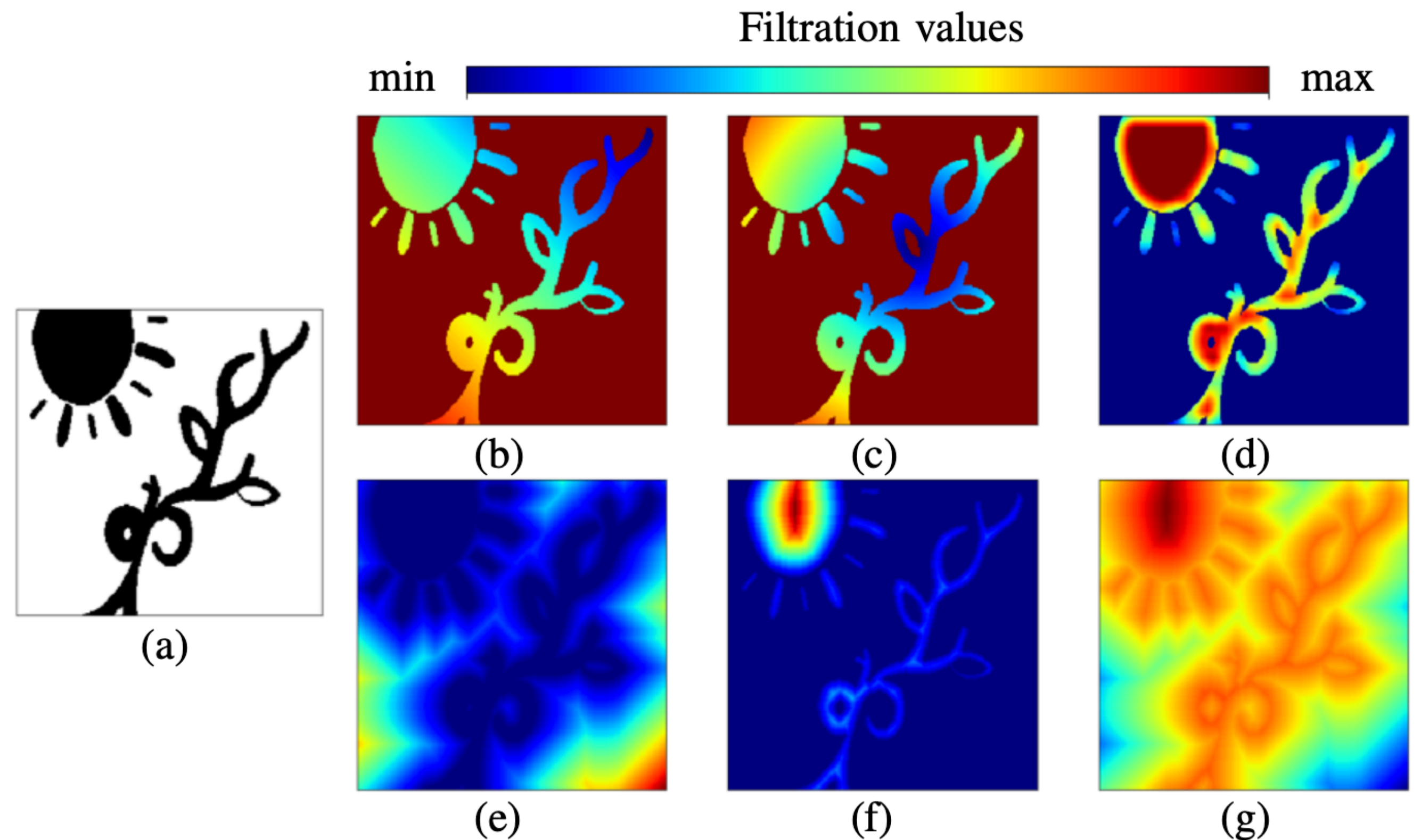
where c is a center point.

Signed Euclidean distance transform (EDT)

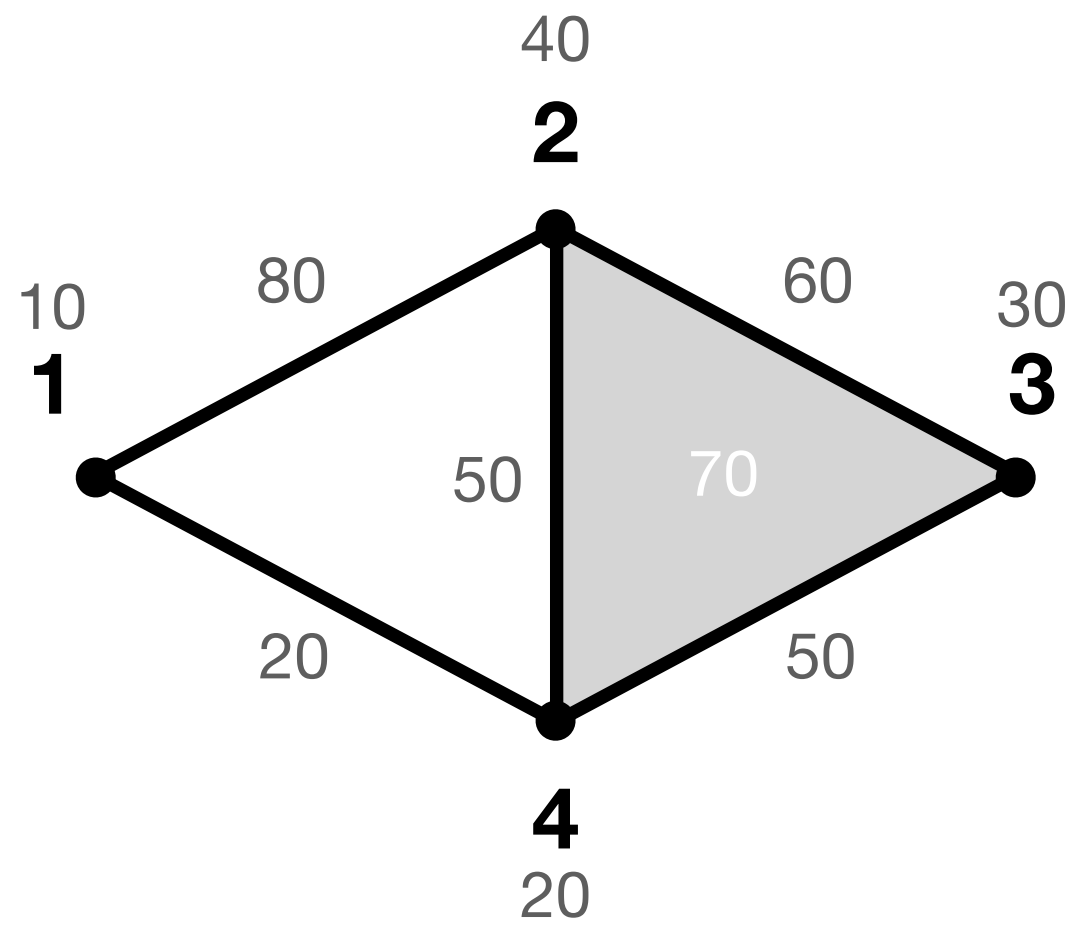
$$f(x) = \begin{cases} d(x, \partial\Omega), & x \in \Omega, \\ -d(x, \partial\Omega), & x \in \Omega^c, \end{cases}$$

where $d(x, \partial\Omega) = \inf_{y \in \partial\Omega} d(x, y)$

$\partial\Omega$ is a boundary of Ω .



Differentiability of the persistent homology



R =

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
1			1							
4			1			1	1			
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

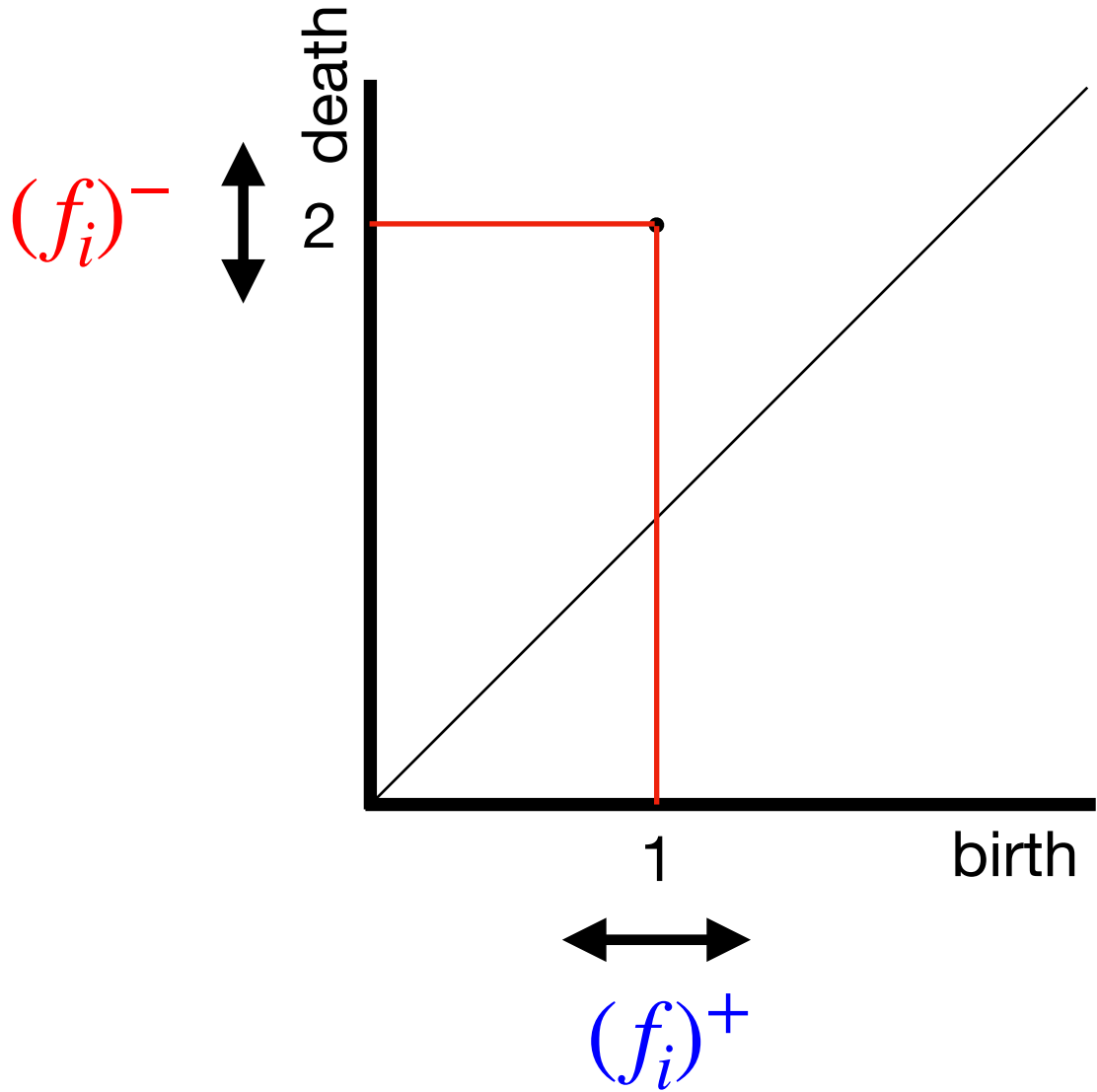
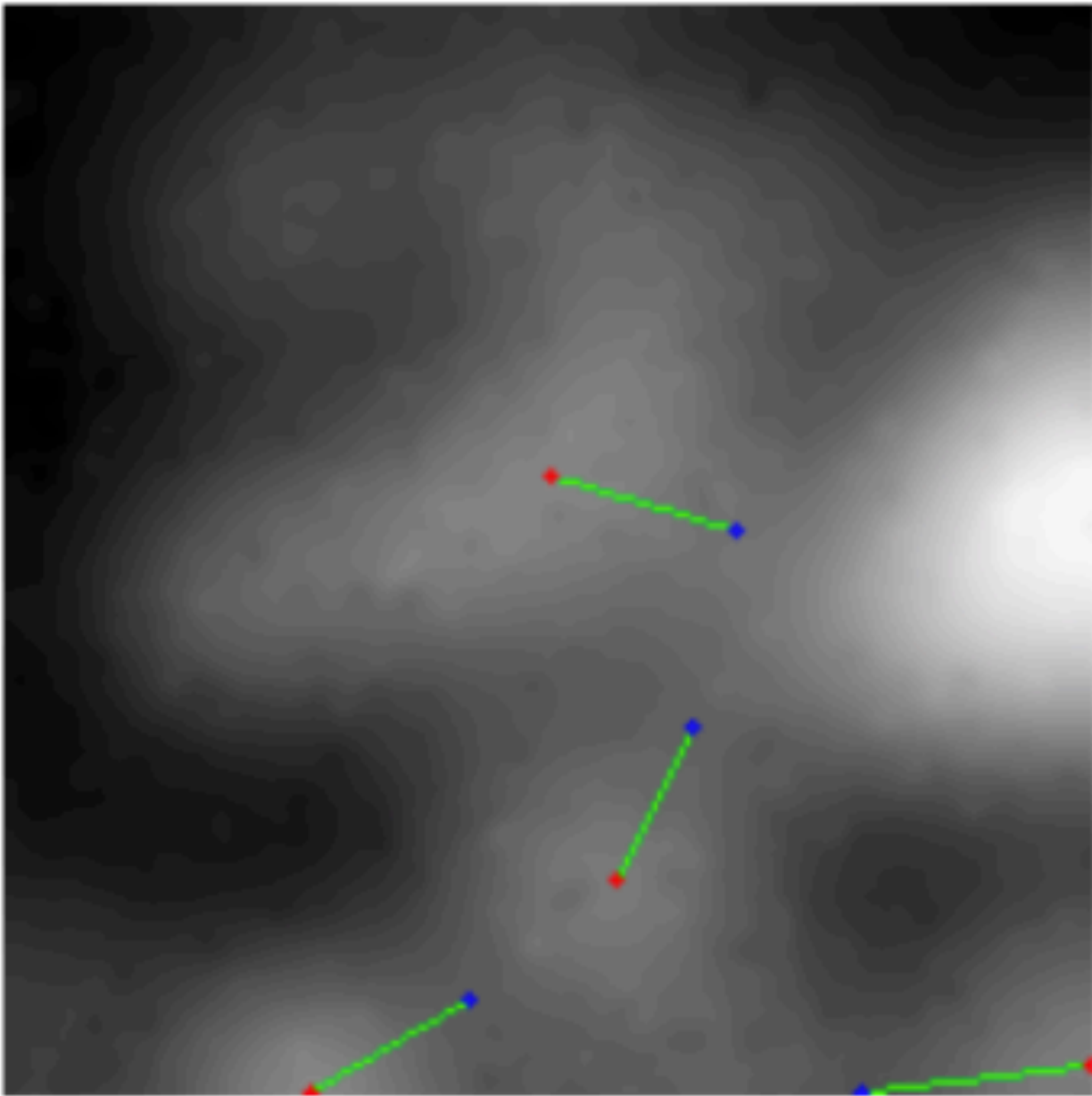
Persistence pairing

(4, 14) 0 (1, ∅) 0
(2, 24) 0 (12, ∅) 1
(3, 34) 0
(23, 234) 1

Persistence diagram

(20, 20) 0 (10, ∅) 0
(40, 50) 0 (80, ∅) 1
(30, 50) 0
(60, 70) 1

Differentiability of the persistent homology

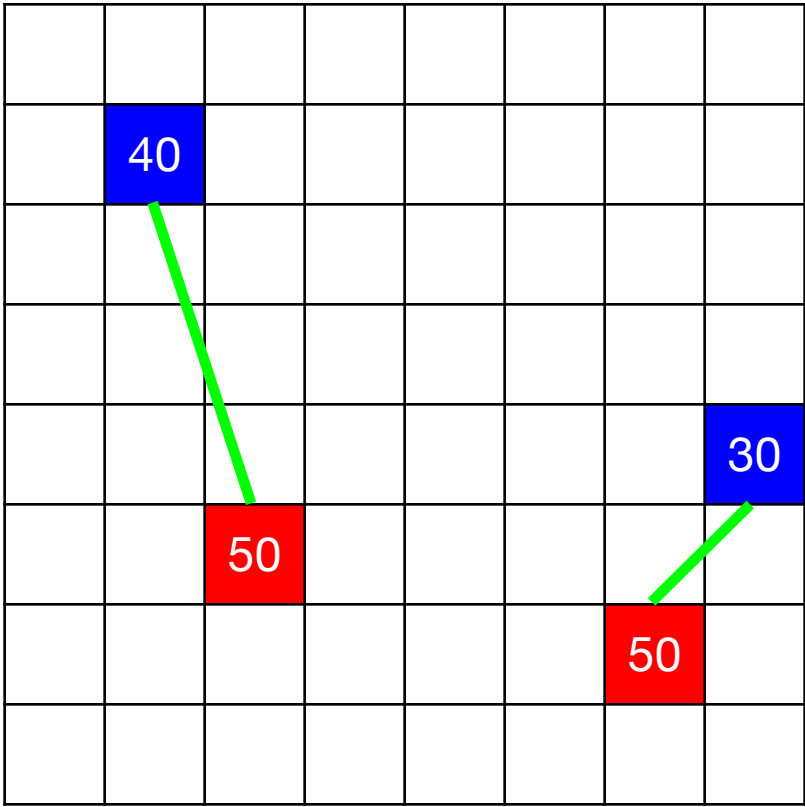


Persistence pairing

(4, 14) 0	(1, <u>0</u>) 0
(2, 24) 0	(12, <u>0</u>) 1
(3, 34) 0	
(23, 234) 1	

Persistence diagram

(20, 20) 0	(10, <u>0</u>) 0
(40, 50) 0	(80, <u>0</u>) 1
(30, 50) 0	
(60, 70) 1	



$$\frac{\partial f_i^+}{\partial b_i} = 1 \quad \frac{\partial f_i^-}{\partial d_i} = -1$$

Software

GUDHI

```
from gudhi.sklearn.cubical_persistence import CubicalPersistence

X = np.random.uniform(size=(100, 100))

pd = CubicalPersistence(newshape=[100, 100])
```

Cubical Ripser

```
import cripsr

X = np.random.uniform(size=(100, 100))

pd = cripsr.computePH(X, maxdim=2)
```

Homcloud

```
import homcloud.interface as hc

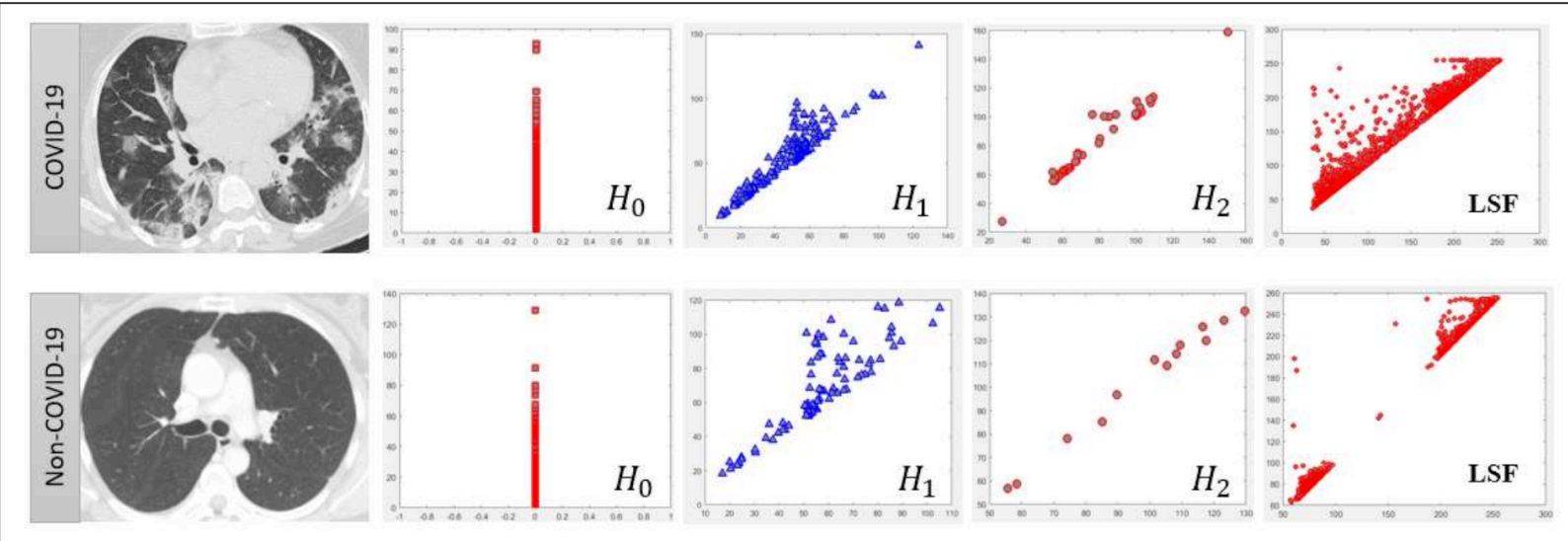
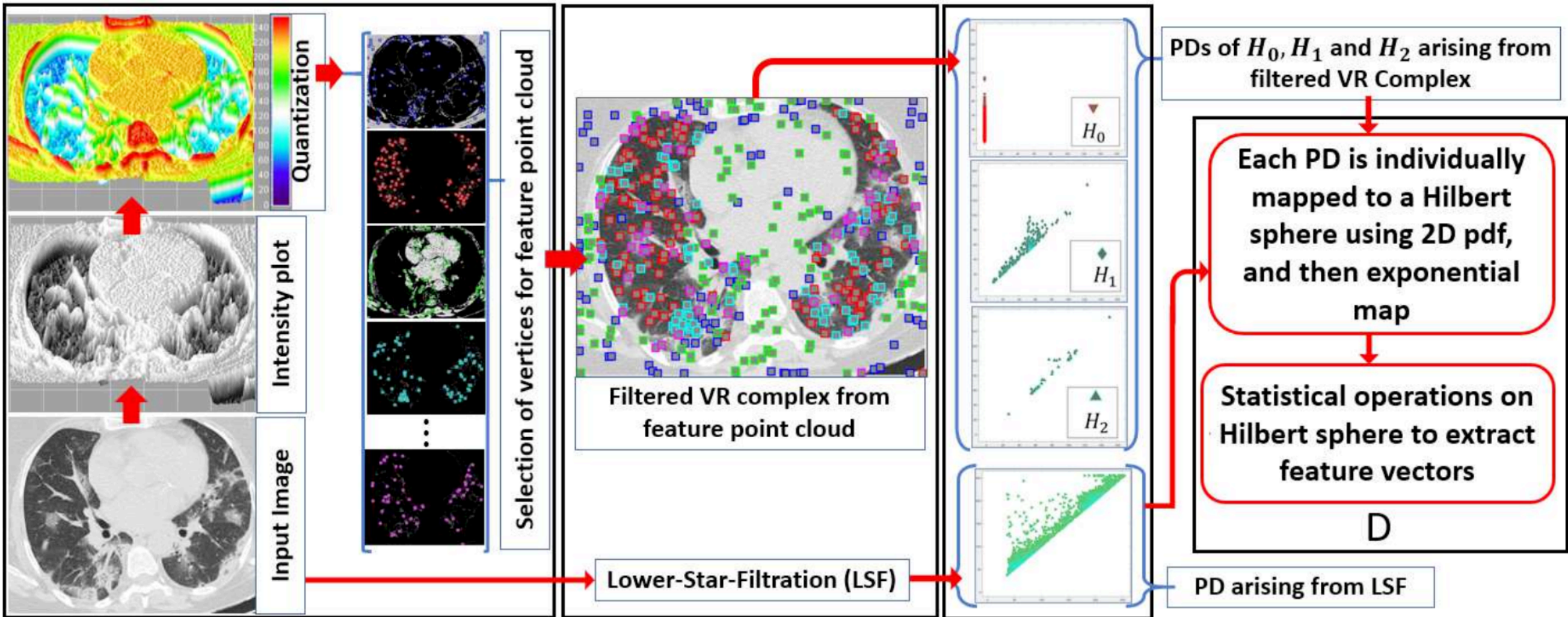
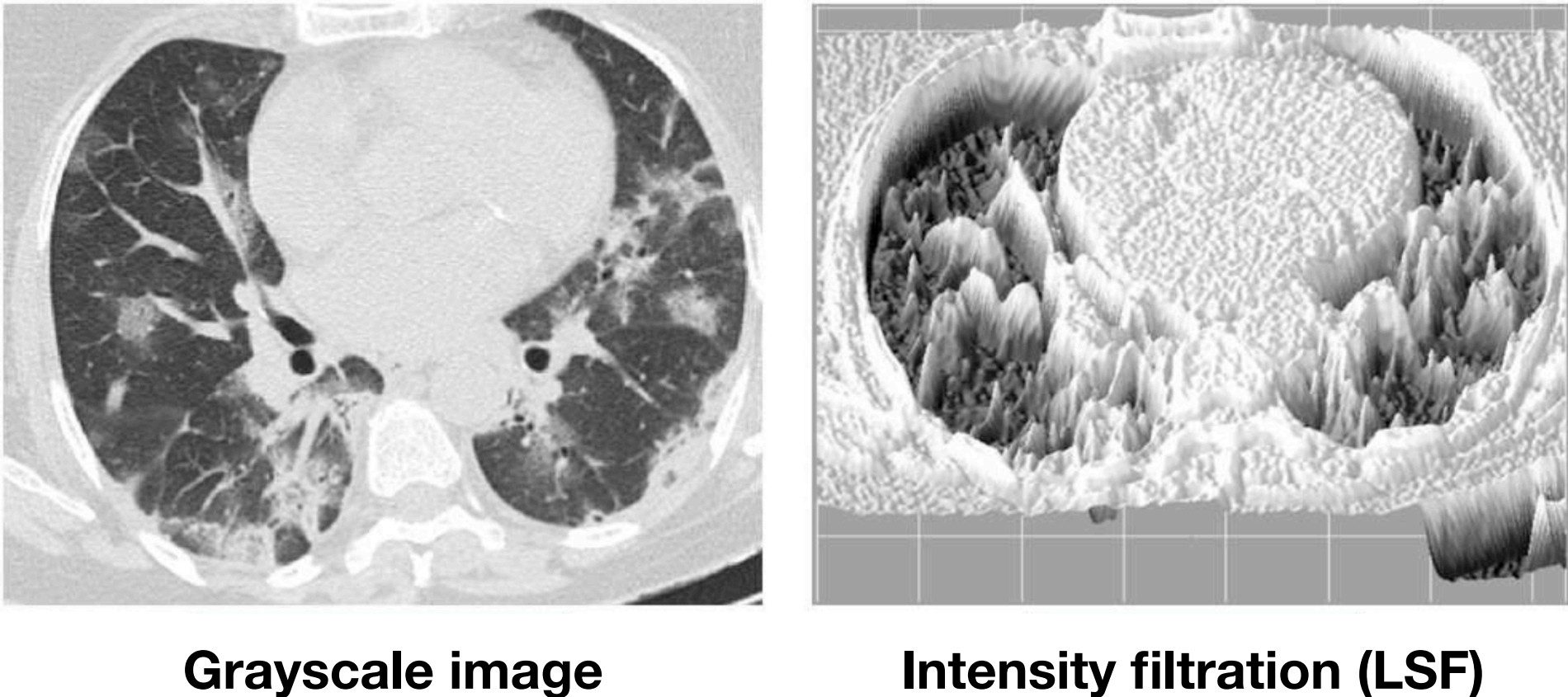
X = np.random.uniform(size=(100, 100))

hc.PDList.from_bitmap_levelset(X, "sublevel", save_to="grayscale.pdgm")

pd0 = hc.PDList("grayscale.pdgm").dth_diagram(0)
pd1 = hc.PDList("grayscale.pdgm").dth_diagram(1)
```


Applications

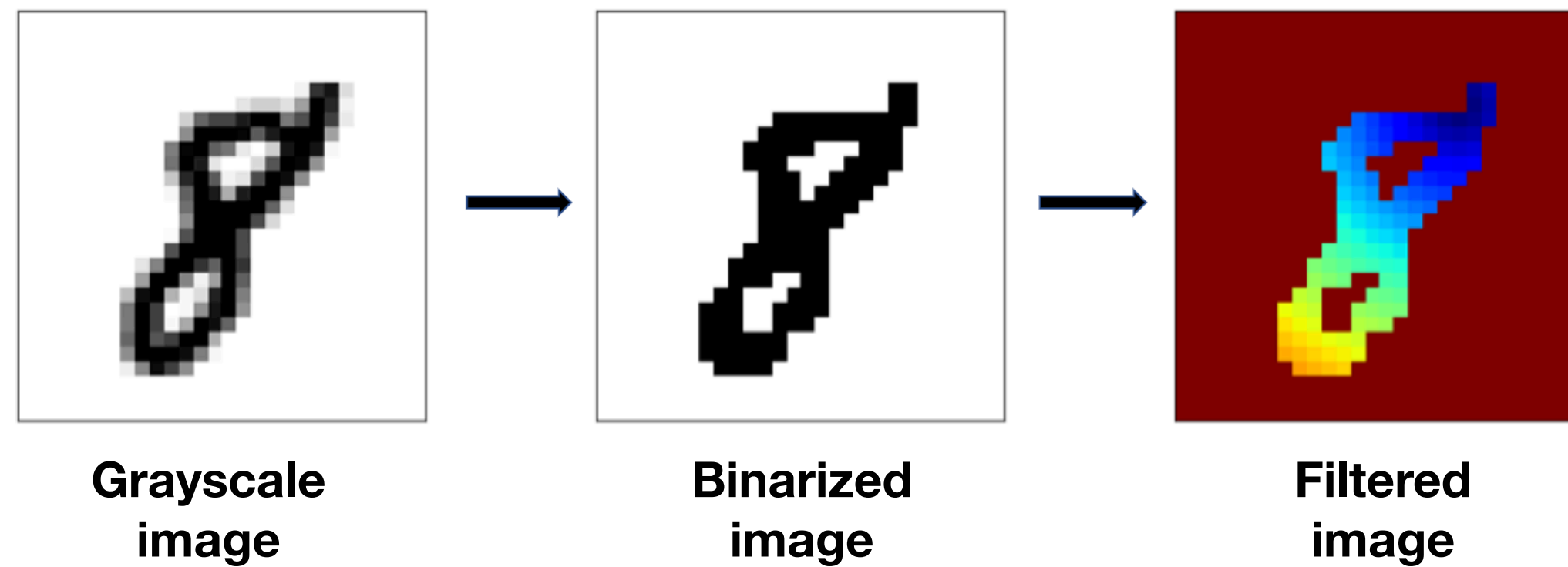
Image classification



Metric \ Top. Feature	Accuracy	Precision	Recall	Specificity	F1 Score
H_0	71.6 ± 2.1	70.3 ± 3.3	74 ± 1.1	69.2 ± 3.7	72 ± 1.93
H_1	68.5 ± 0.5	67.1 ± 2.6	72.6 ± 7.7	64.4 ± 8.1	69.5 ± 32.3
H_2	64.8 ± 2.1	63.5 ± 1.8	67.7 ± 4.7	61.9 ± 2.2	65.5 ± 2.5
LSF	97.5 ± 0.6	97 ± 1	98 ± 0.6	97 ± 1	97.5 ± 0.6

Applications

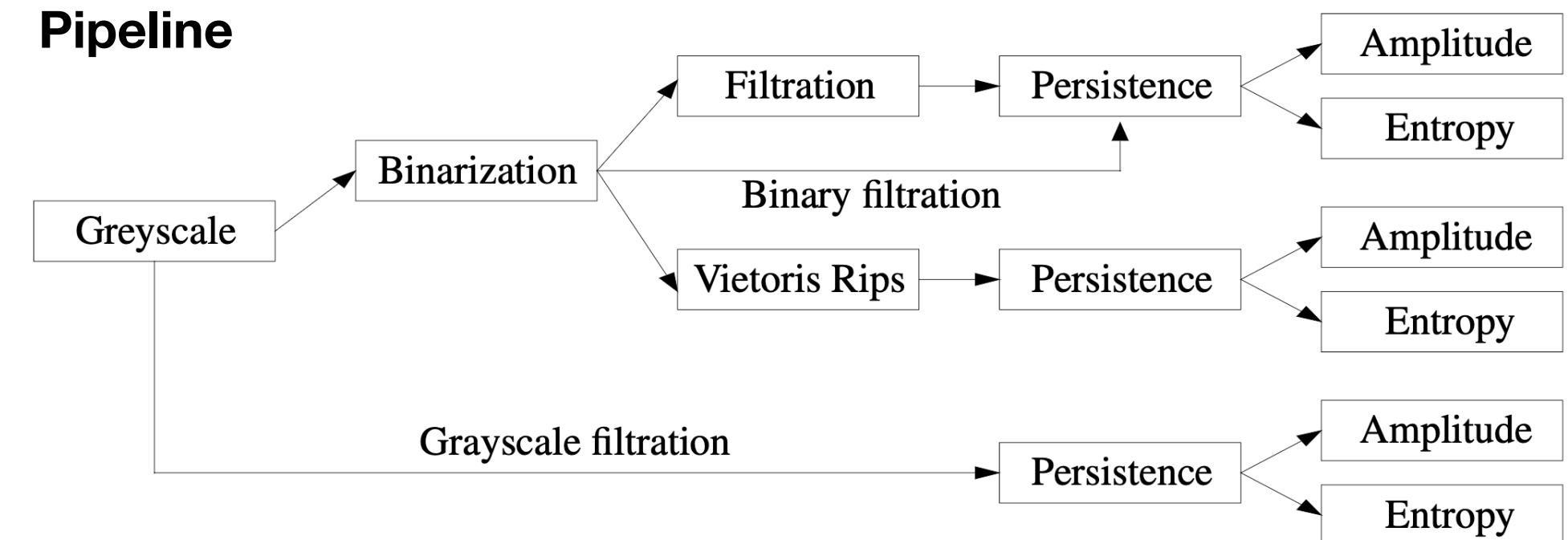
Image classification



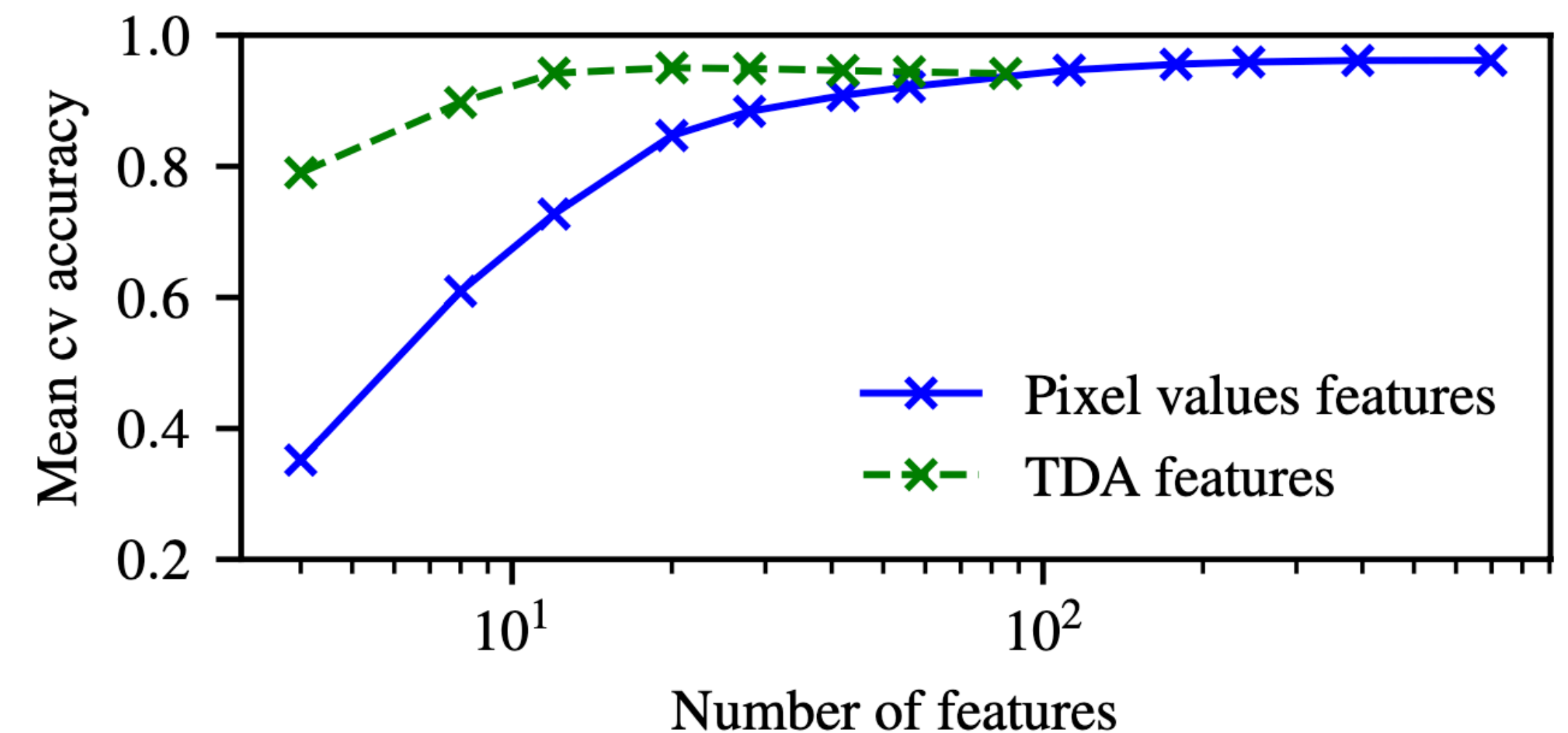
Filtrations

- direction filtration
- radial filtration
- signed EDT filtration
- density filtration

Pipeline



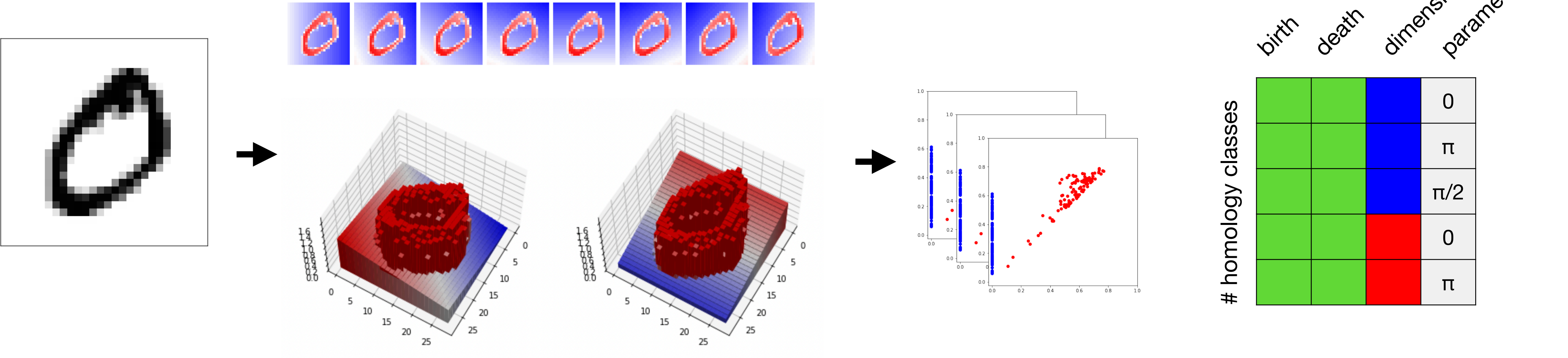
Results



Persistent homology transform

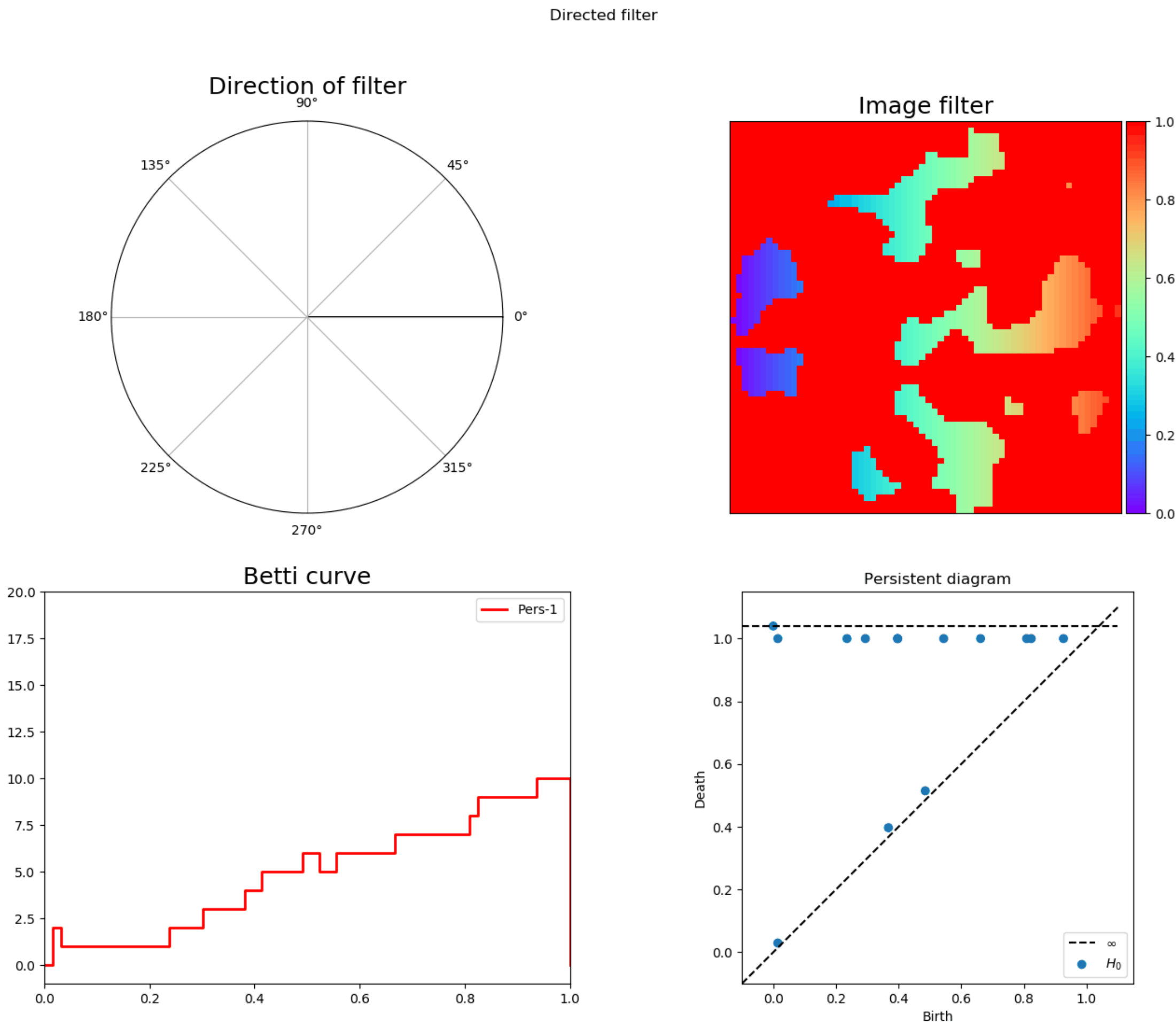
Persistent homology transform [8] maps a topological space X along with a family of parameterized filtration functions $\{f_\theta\}_{\theta \in \Theta}$, such that to a family of persistent diagrams.

$$F : X \times \Theta \rightarrow \mathcal{D}_\Theta \tag{1}$$



Applications

Image classification



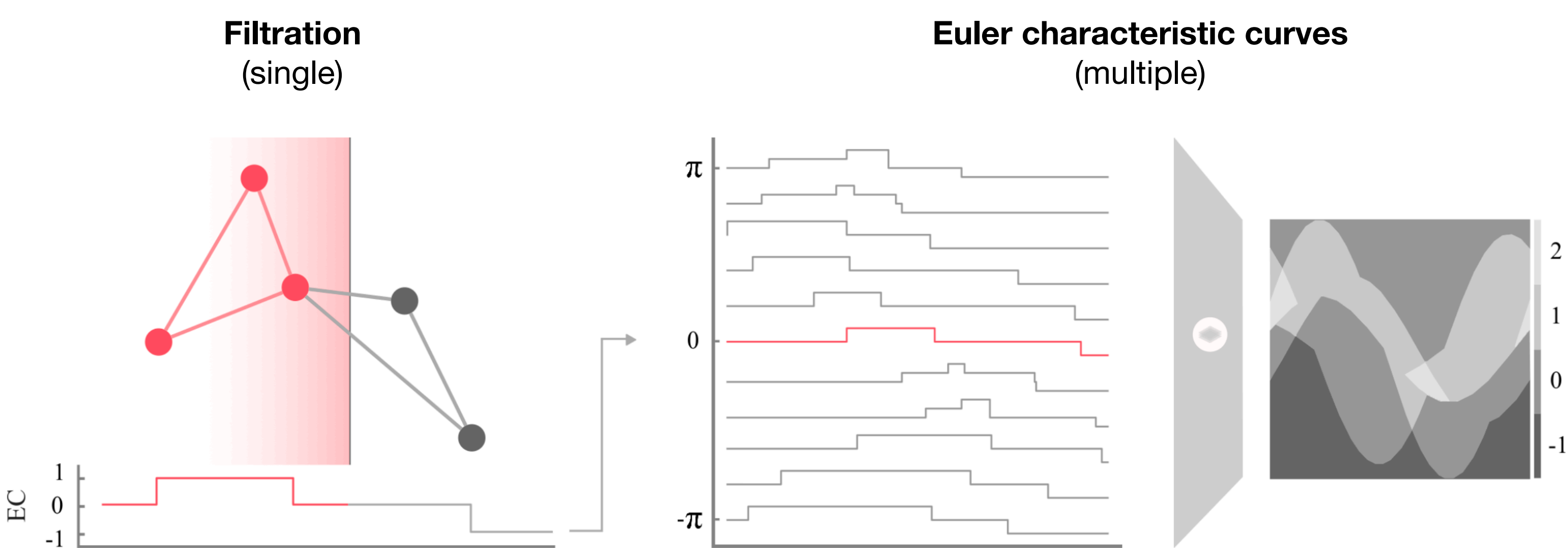
	Accuracy
Fixed filter	0,7249

Classification accuracy

	# filters	
	4	10
Fixed filters	0,6631	0,7493
Fixed filters similarity matrix	0,7512	0,9220
Trainable filters similarity matrix	0,7611	0,9330

Classification accuracy

Euler characteristic transform



Euler characteristic

$$\chi = \sum_{i=0}^d (-1)^i \beta_i$$

Poincaré-Euler

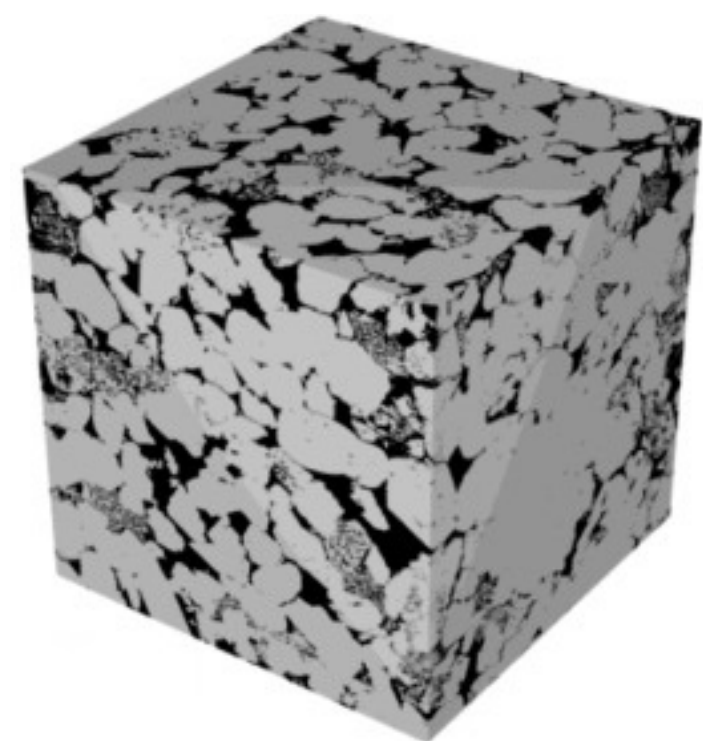
$$\chi = \sum_{i=0}^d (-1)^i \# \sigma^{(i)}$$

Euler characteristic curve

$$\chi(t) = \sum_{i=0}^d (-1)^i \# \sigma^{(i)}(t)$$

Applications

Permeability prediction



Porous media

Physical properties

- permeability
- conductivity
- strength
- etc.

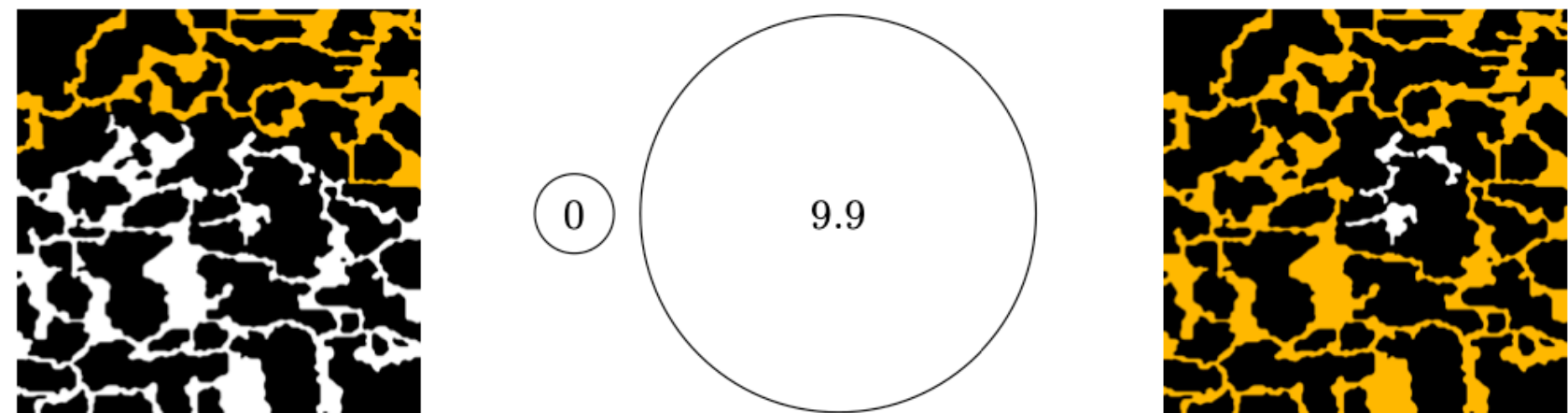


Рис. 6. Для образца А поровое пространство непроницаемо. Максимальный радиус шара проходящего через пространство пор образца Б – 9.9.

Permeability		Filtration		
		EDT	Vertical	Horizontal
Model [24,18]	Mean	7.76 ± 0.00	8.04 ± 0.00	8.28 ± 0.00
	Vertical	14.76 ± 0.00	10.48 ± 0.00	–
	Horizontal	12.01 ± 0.00	–	9.45 ± 0.00

Media	Direction	Filtration	
		EDT	Bilateral
Bentheimer	Mean	0.527 ± 0.023	0.412 ± 0.015
	Top-bottom	0.613 ± 0.029	0.411 ± 0.016
	Left-right	0.846 ± 0.025	0.570 ± 0.024
	Front-back	0.699 ± 0.019	0.575 ± 0.026
Doddington	Mean	1.295 ± 0.037	0.956 ± 0.042
	Top-bottom	1.401 ± 0.057	0.944 ± 0.042
	Left-right	1.835 ± 0.069	1.250 ± 0.032
	Front-back	1.435 ± 0.053	1.044 ± 0.049
Ketton	Mean	1.077 ± 0.046	0.763 ± 0.031
	Top-bottom	1.189 ± 0.045	1.059 ± 0.045
	Left-right	1.459 ± 0.055	0.889 ± 0.028
	Front-back	1.289 ± 0.051	0.958 ± 0.048

Permeability prediction error, % for 2D model and 3D real data.

Applications

Topology-denoising Autoencoder

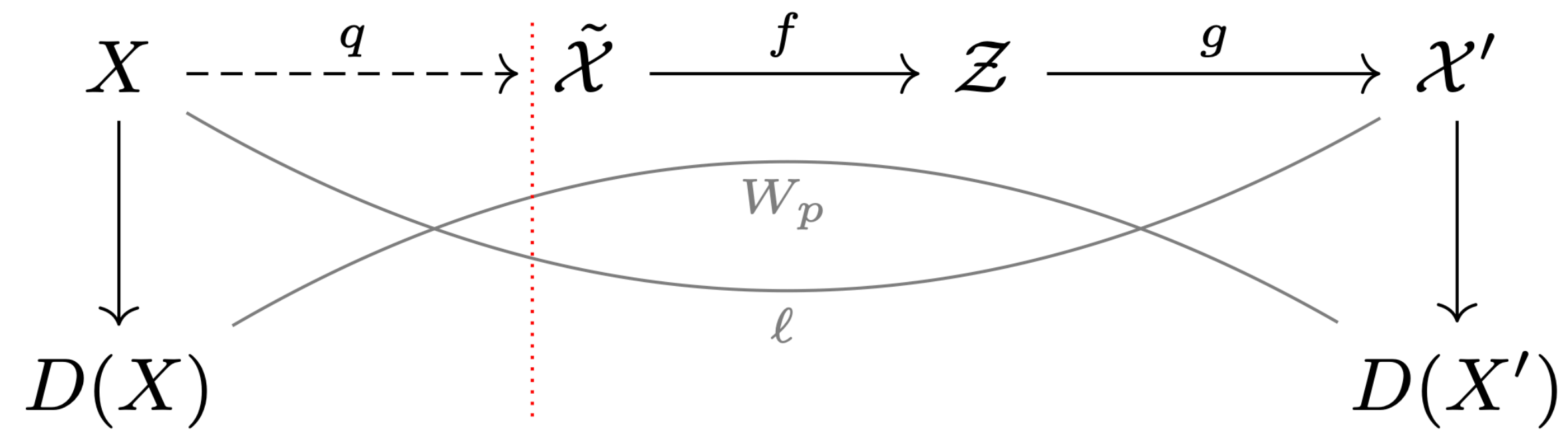
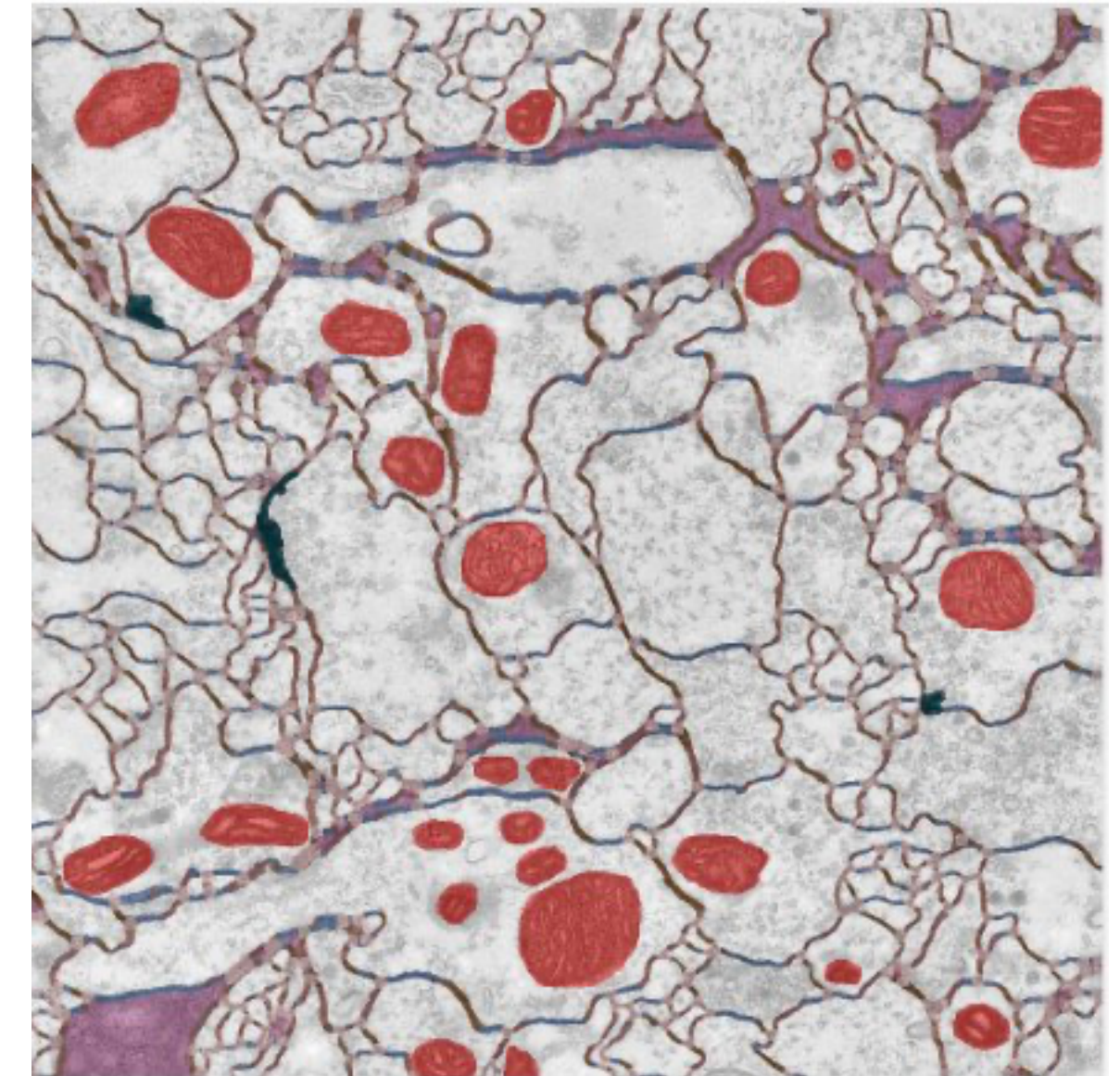


Fig. 3. Topology-denoising autoencoder (TDAE)



Applications

Topology-denoising Autoencoder

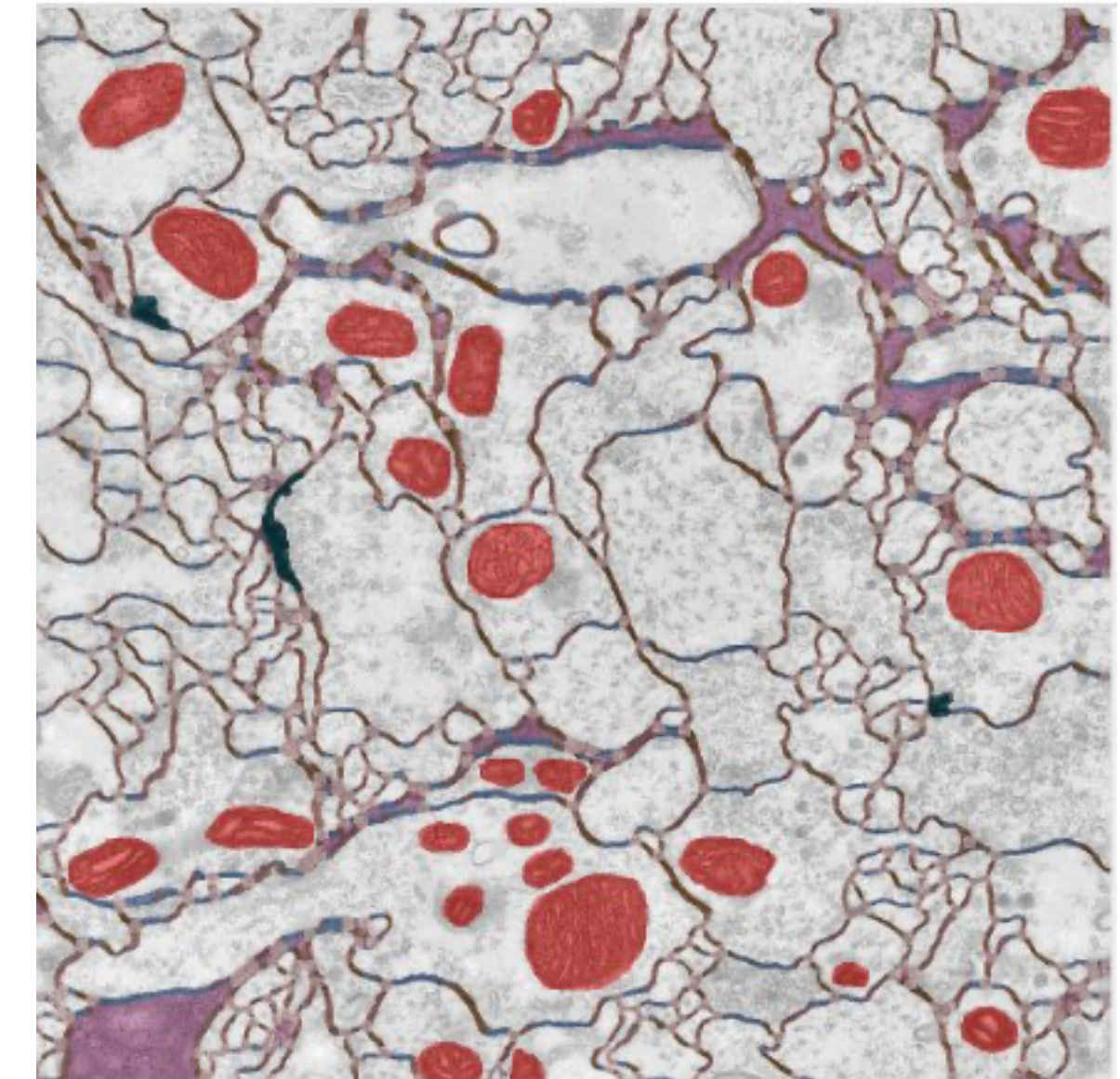
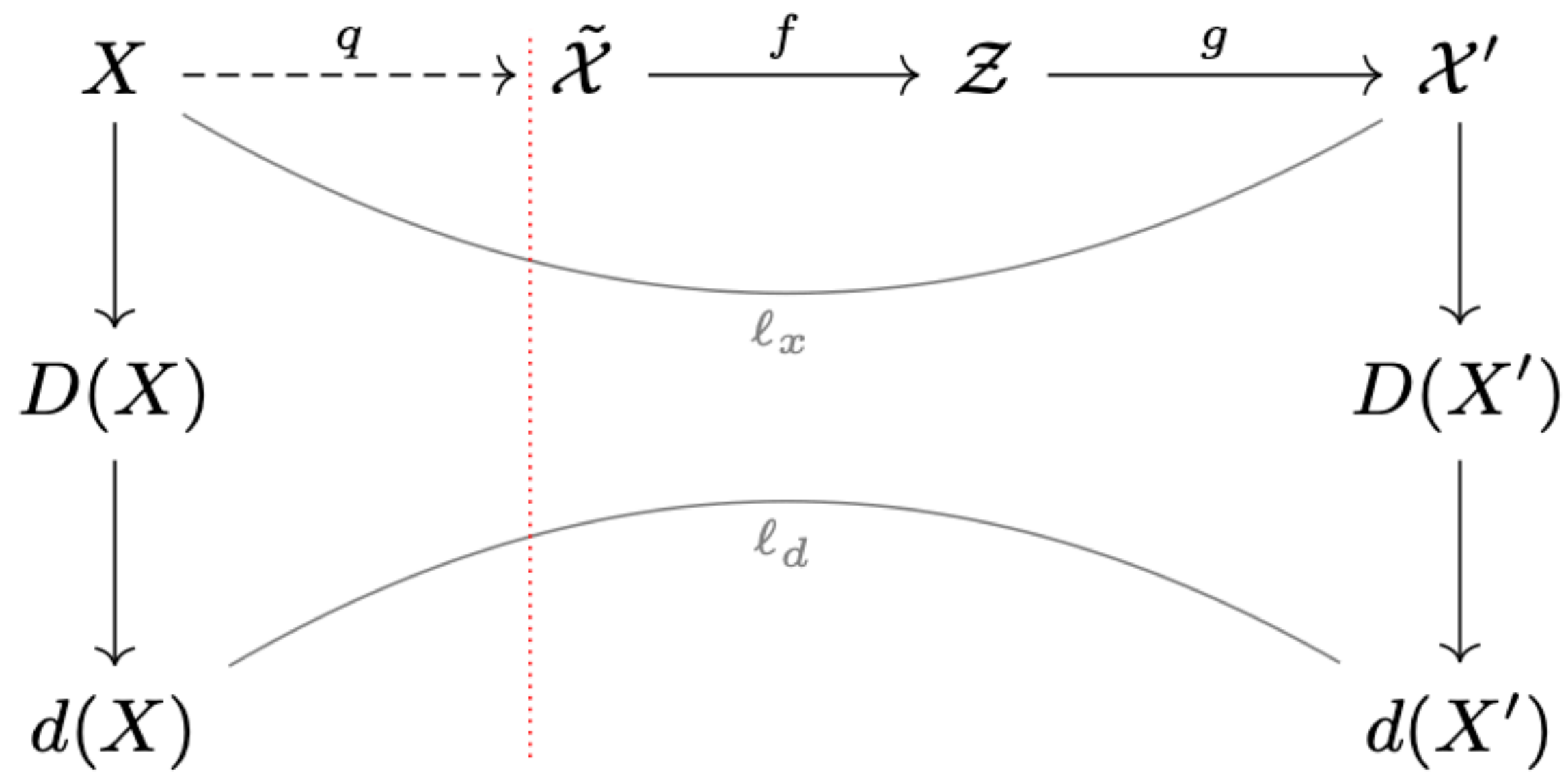
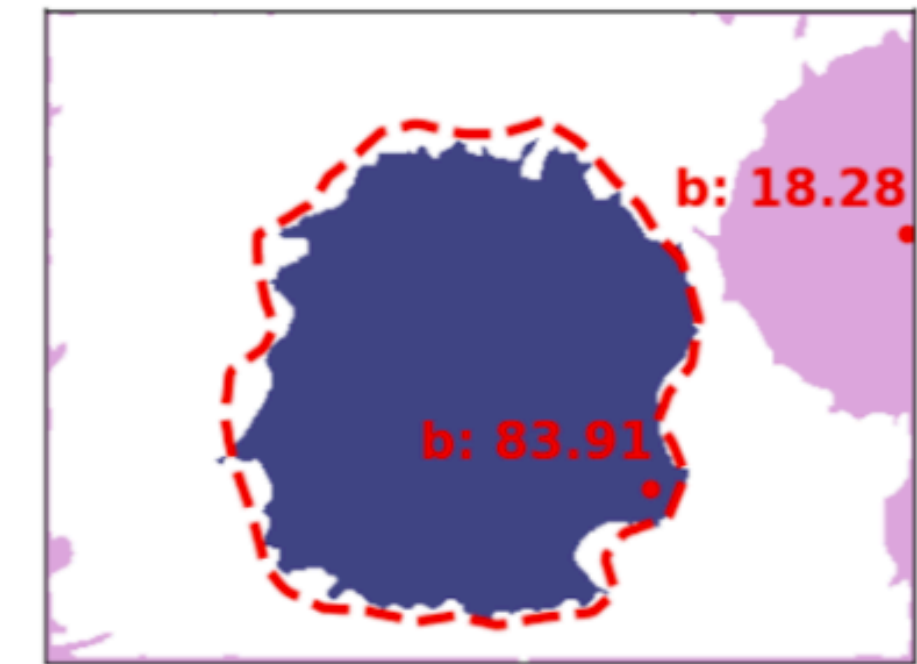
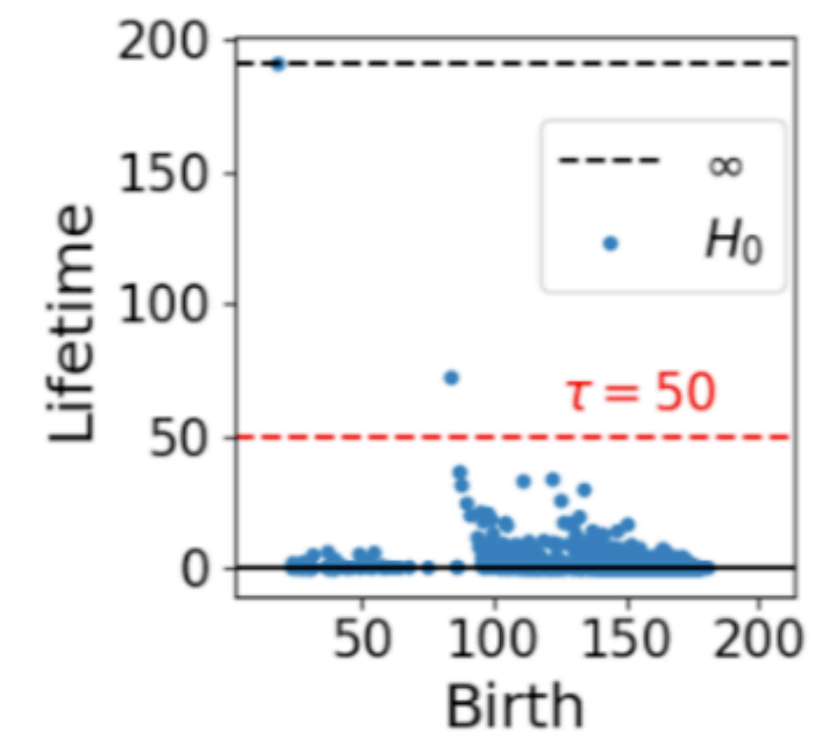
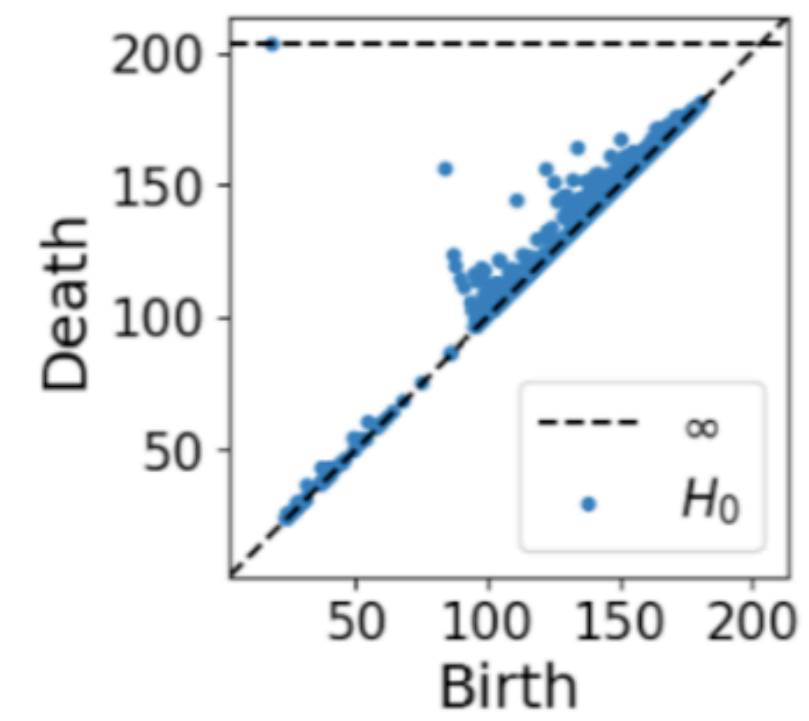
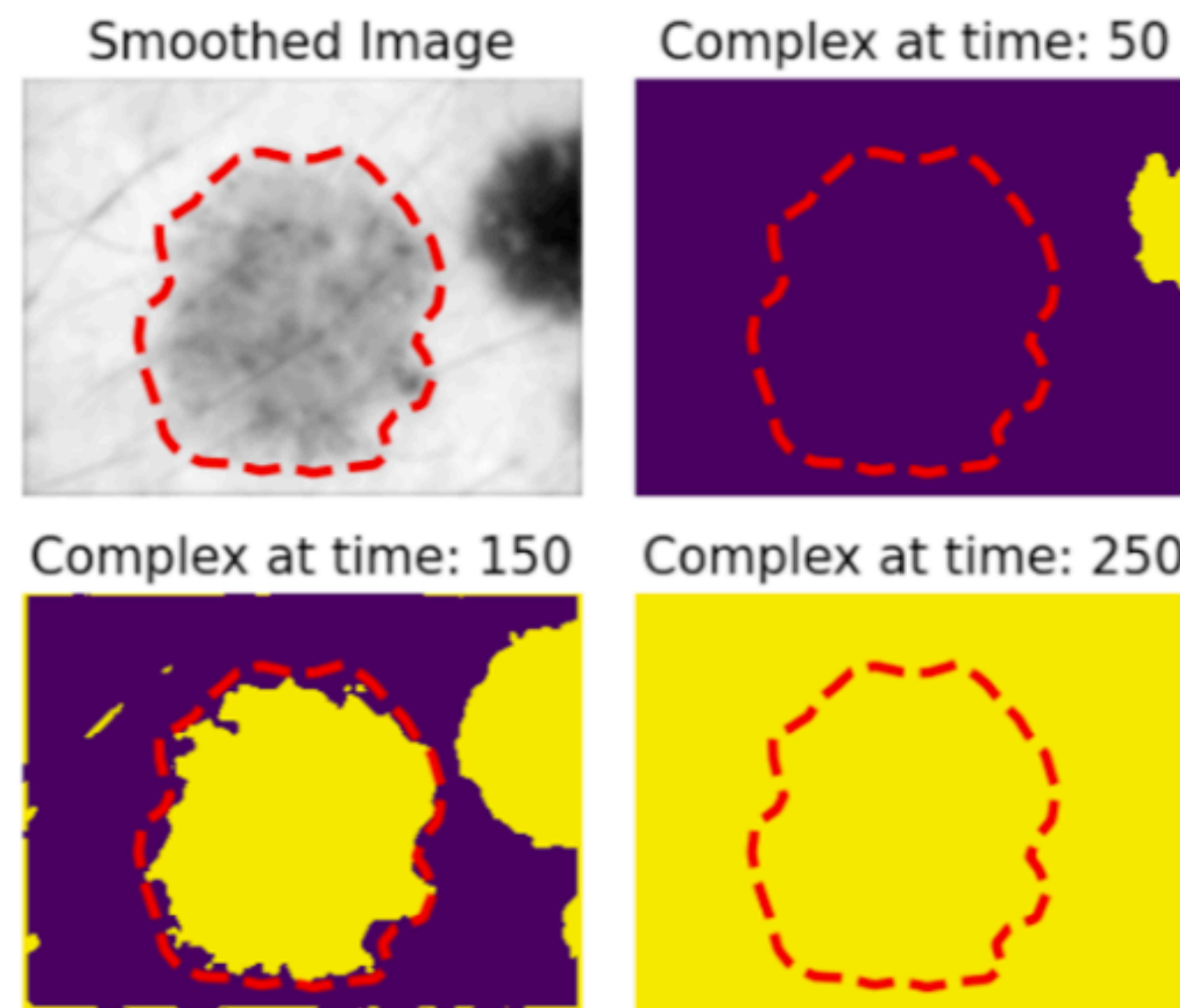


Fig. 4. Topology-denoising autoencoder, variant (TDAE)

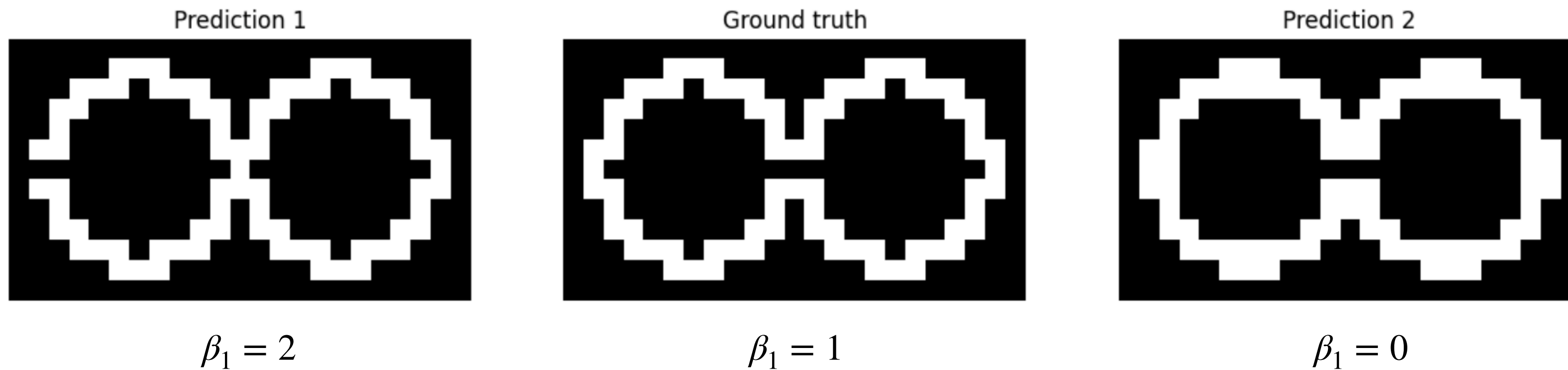
Applications

Topology-aware segmentation



Applications

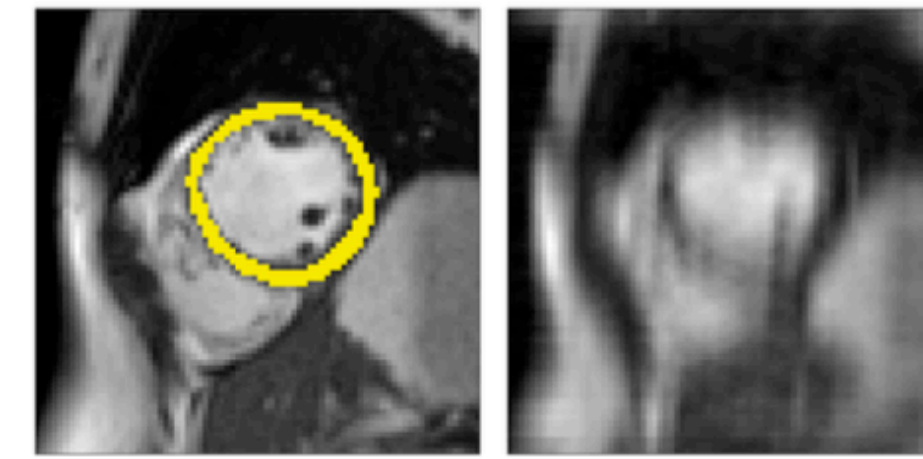
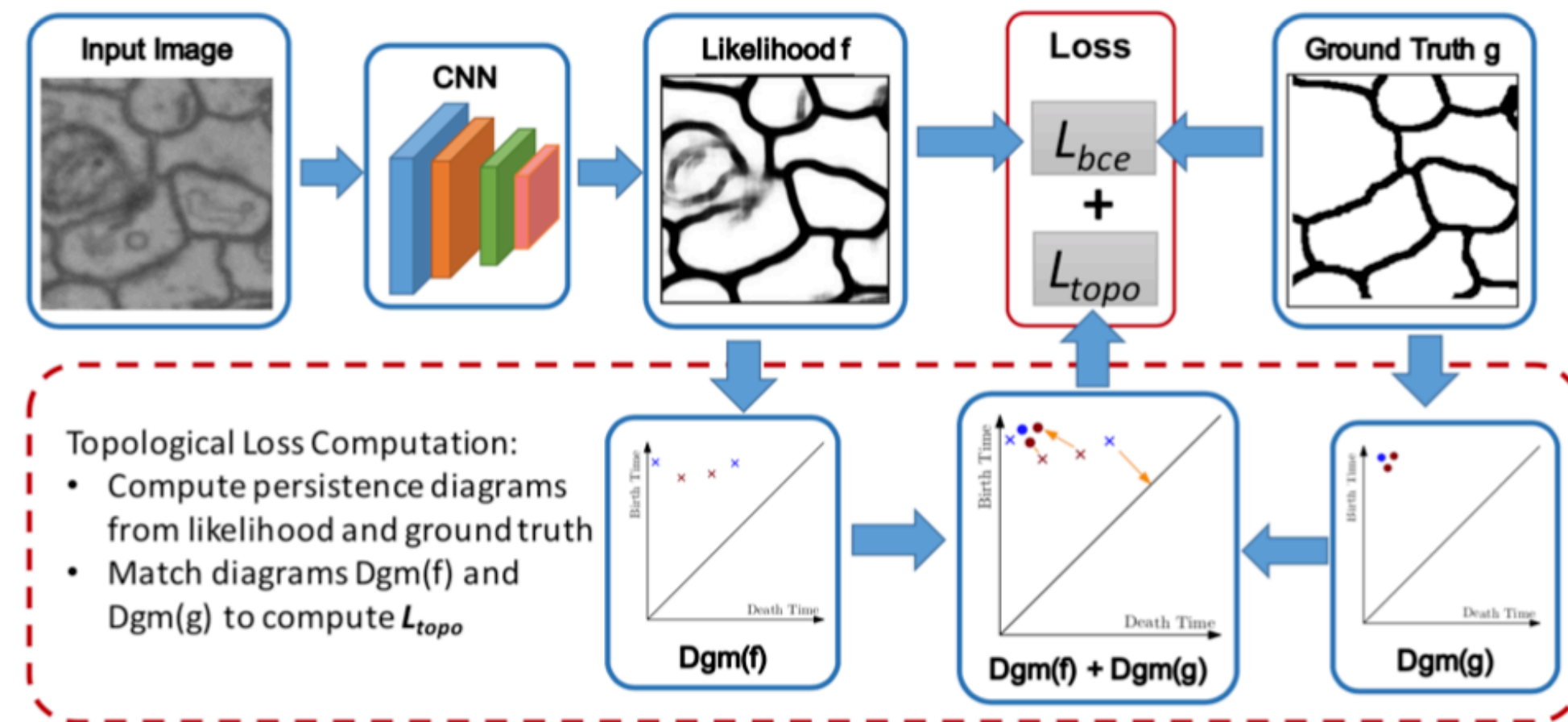
Topology-aware segmentation



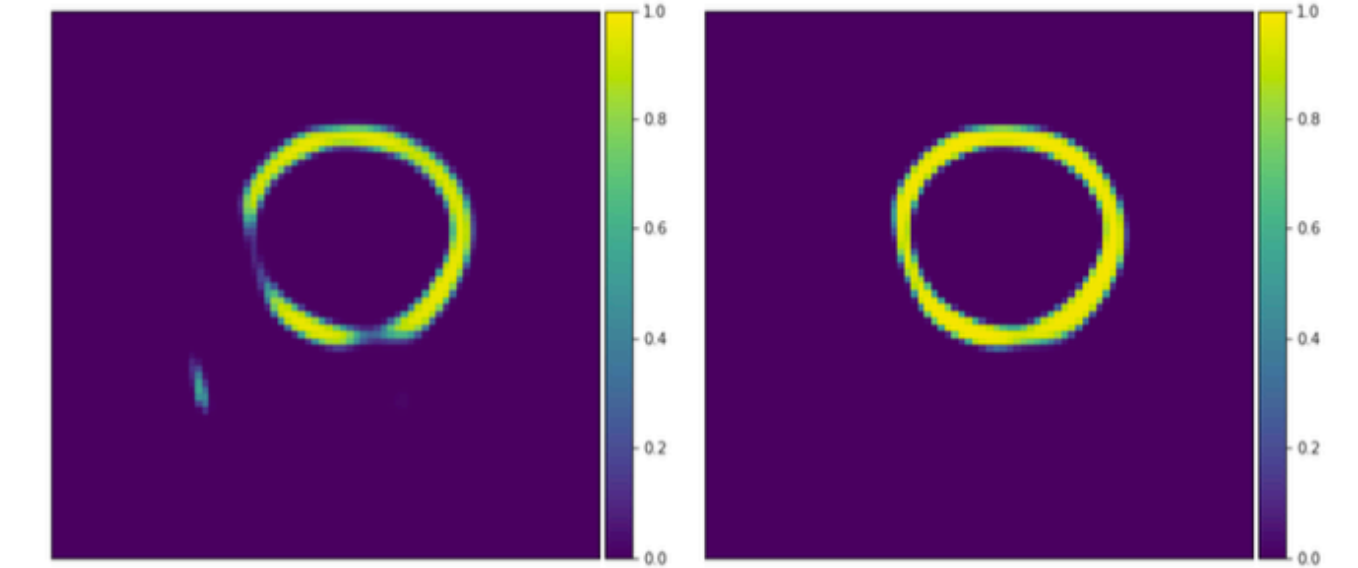
Dice score:
Prediction 1: 0.038
Prediction 2: 0.055

Applications

Topology-aware segmentation



(a) Left: Uncorrupted image and ground-truth segmentation. Right: Corrupted image, the input to the network.



(b) Left: The predicted segmentation from the network trained only with supervised learning. Right: The predicted segmentation from the network trained in a semi-supervised manner, incorporating the topological prior.

General form $L(g, \hat{g}) = L_{CE}(g, \hat{g}) + \lambda L_{TOPO}(g, \hat{g})$

Topological regularizer (Hu et al.)

$$L_{TOPO}(g, \hat{g}) = d_{W_2}(D(g), D(\hat{g}))$$

Topological regularizer (Clough et al.) – “desired Betti numbers” β_k^*

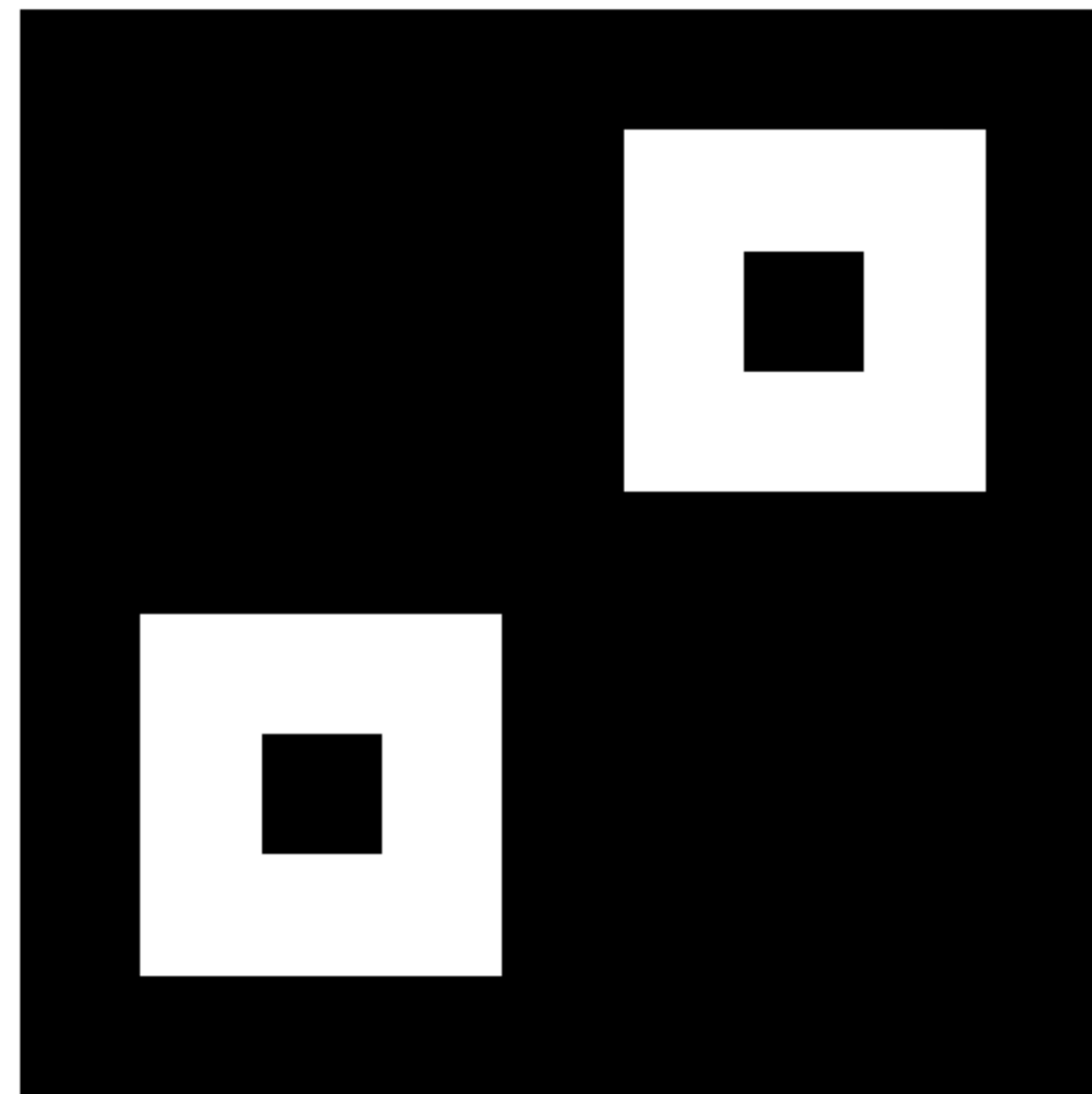
$$L_{TOPO}(g, \hat{g}) = \sum_k L_k(\beta_k^*), \text{ where}$$

$$L_k(\beta_k^*) = \sum_{\ell=1}^{\beta_k^*} (1 - (d_k, \ell - b_{k,\ell}))^2 + \sum_{\ell=\beta_k^*+1}^{\infty} (d_k, \ell - b_{k,\ell})^2.$$

Applications

Localized topology-aware segmentation

Ground truth

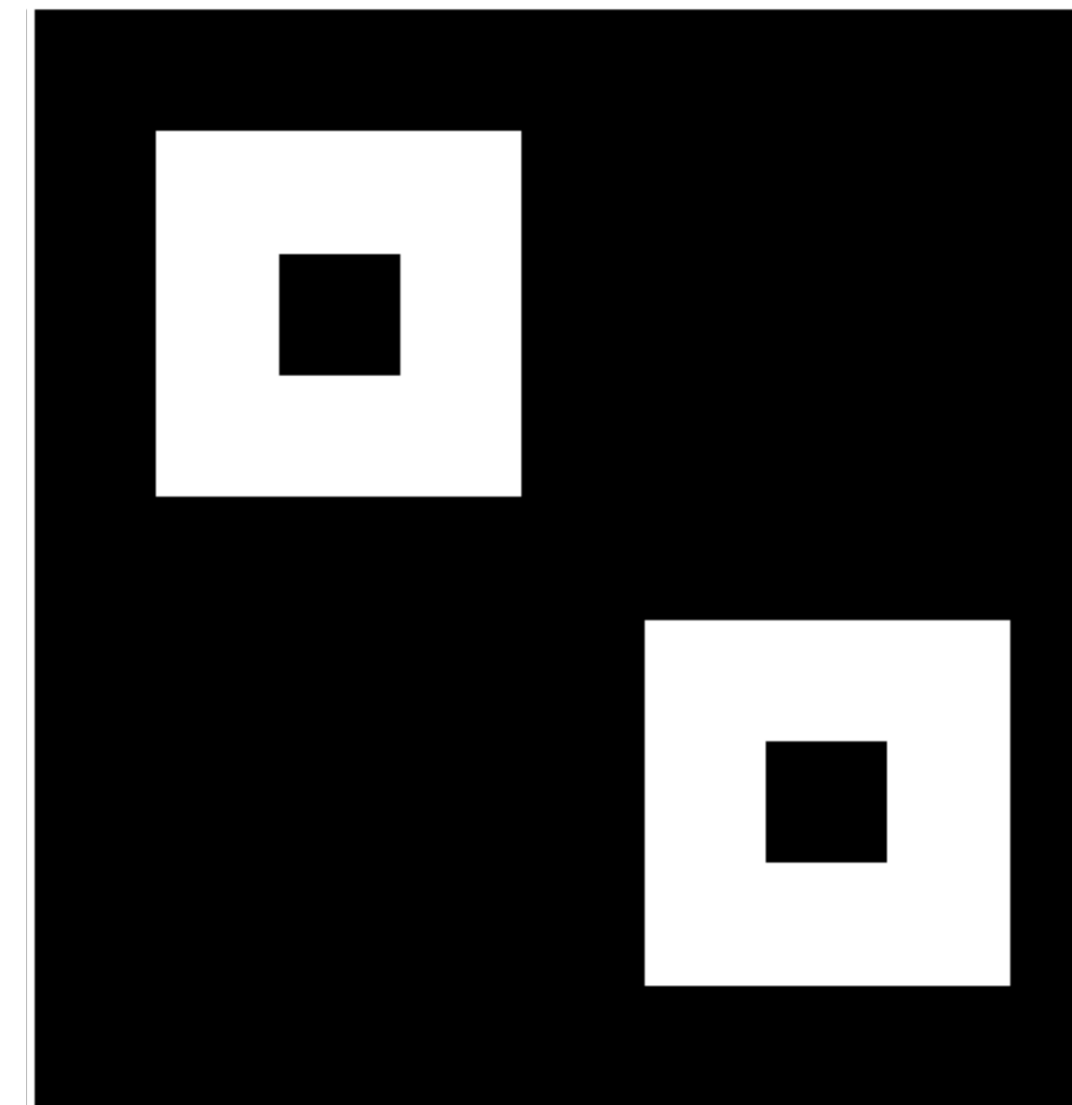


$$\beta_0 = 2$$

$$\beta_1 = 2$$

$$\ell(\beta_0, \hat{\beta}_0) = 0 \quad \ell(\beta_1, \hat{\beta}_1) = 0$$

Prediction



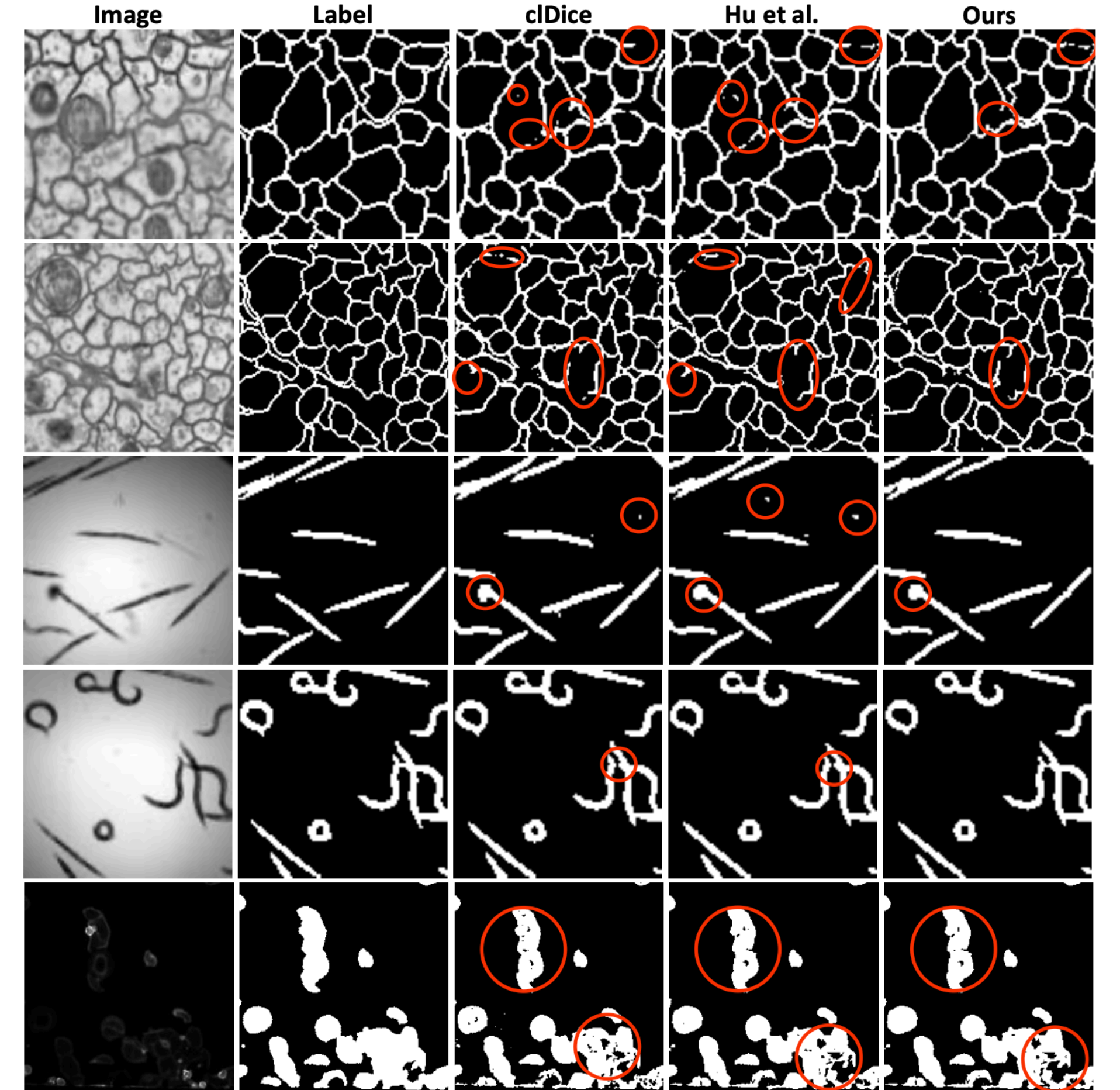
$$\hat{\beta}_0 = 2$$

$$\hat{\beta}_1 = 2$$

$$\mu(\beta_0, \hat{\beta}_0) = 2 \quad \mu(\beta_1, \hat{\beta}_1) = 2$$

Localized topology-aware segmentation

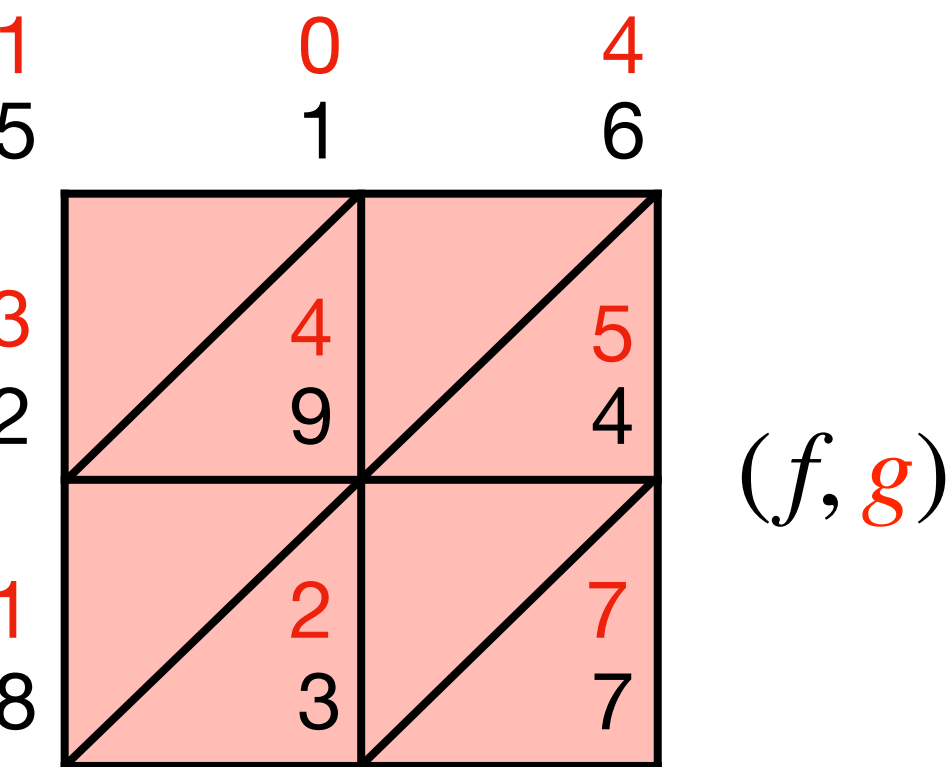
	Loss	Dice $\uparrow$	clDice $\uparrow$	Acc. $\uparrow$	$\mu^{\text{err}} \downarrow$	$\mu_0^{\text{err}} \downarrow$	$\mu_1^{\text{err}} \downarrow$	$\beta^{\text{err}} \downarrow$	$\beta_0^{\text{err}} \downarrow$	$\beta_1^{\text{err}} \downarrow$
CREMI	Dice	0.894	0.939	0.959	149.64	39.68	109.96	114.12	39.12	75.00
	clDice	0.879	0.944	0.952	147.04	34.36	112.68	103.92	33.64	70.28
	Hu et al.	0.888	0.935	0.957	162.48	44.24	118.24	118.16	43.68	74.48
	Ours	0.893	0.941	0.959	129.80	31.00	98.80	79.16	30.36	48.80
Roads	Dice	0.663	0.698	0.974	117.80	87.04	30.76	113.96	86.54	27.42
	clDice	0.668	0.704	0.975	131.00	102.08	28.92	125.83	101.67	24.17
	Hu et al.	0.674	0.712	0.974	101.00	73.04	27.96	95.83	72.54	23.29
	Ours	0.663	0.713	0.972	83.00	56.30	26.70	75.08	55.79	19.29
synMnist	Dice	0.871	0.907	0.962	3.70	1.96	1.74	2.590	1.674	0.916
	clDice	0.875	0.921	0.963	2.54	0.87	1.67	1.640	0.700	0.940
	Hu et al.	0.866	0.915	0.960	2.85	1.00	1.85	1.802	0.764	1.038
	Ours	0.849	0.915	0.954	2.28	0.53	1.75	1.348	0.426	0.922
Elegans	Dice	0.922	0.959	0.984	4.10	2.60	1.50	2.60	1.40	1.20
	clDice	0.917	0.964	0.982	3.90	2.20	1.70	2.20	1.20	1.00
	Hu et al.	0.921	0.959	0.984	4.30	2.84	1.45	2.50	1.35	1.15
	Ours	0.919	0.960	0.983	3.40	2.10	1.30	1.90	0.80	1.10
Colon	Dice	0.899	0.863	0.970	44.26	21.76	22.50	33.75	13.75	20.00
	clDice	0.907	0.871	0.974	47.26	18.76	28.50	37.75	11.75	26.00
	Hu et al.	0.902	0.876	0.972	34.50	15.50	19.00	22.00	7.00	15.00
	Ours	0.907	0.871	0.975	32.00	14.26	17.76	21.50	6.25	15.25
Buildings	Dice	0.623	0.672	0.934	572.44	551.00	21.46	162.95	151.70	11.25
	clDice	0.632	0.693	0.931	571.20	535.96	35.26	175.50	155.05	20.45
	Hu et al.	0.625	0.677	0.934	556.60	537.50	19.10	181.10	169.60	11.50
	Ours	0.625	0.685	0.937	489.16	471.26	17.90	118.45	107.75	10.70



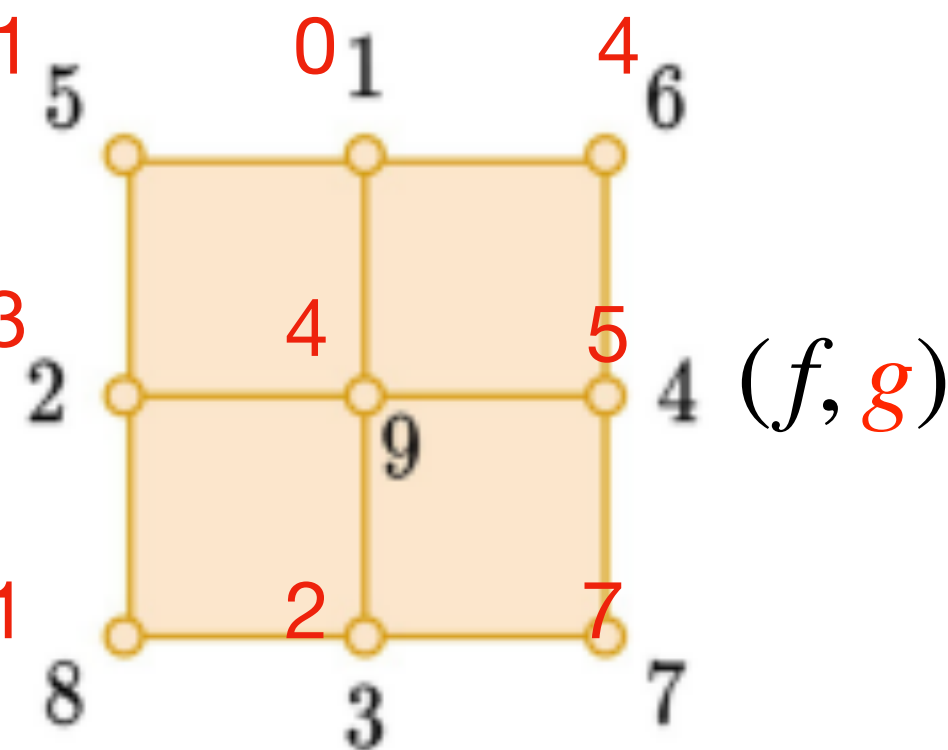
Scalar function topology divergence

$$d \left(\begin{array}{c} \begin{bmatrix} 5 & 1 & 6 \\ 2 & 9 & 4 \\ 8 & 3 & 7 \end{bmatrix} \\ f : I \rightarrow \mathbb{R}_{\geq 0} \end{array} , \begin{array}{c} \begin{bmatrix} 1 & 0 & 4 \\ 3 & 4 & 5 \\ 1 & 2 & 7 \end{bmatrix} \\ g : I \rightarrow \mathbb{R}_{\geq 0} \end{array} \right)$$

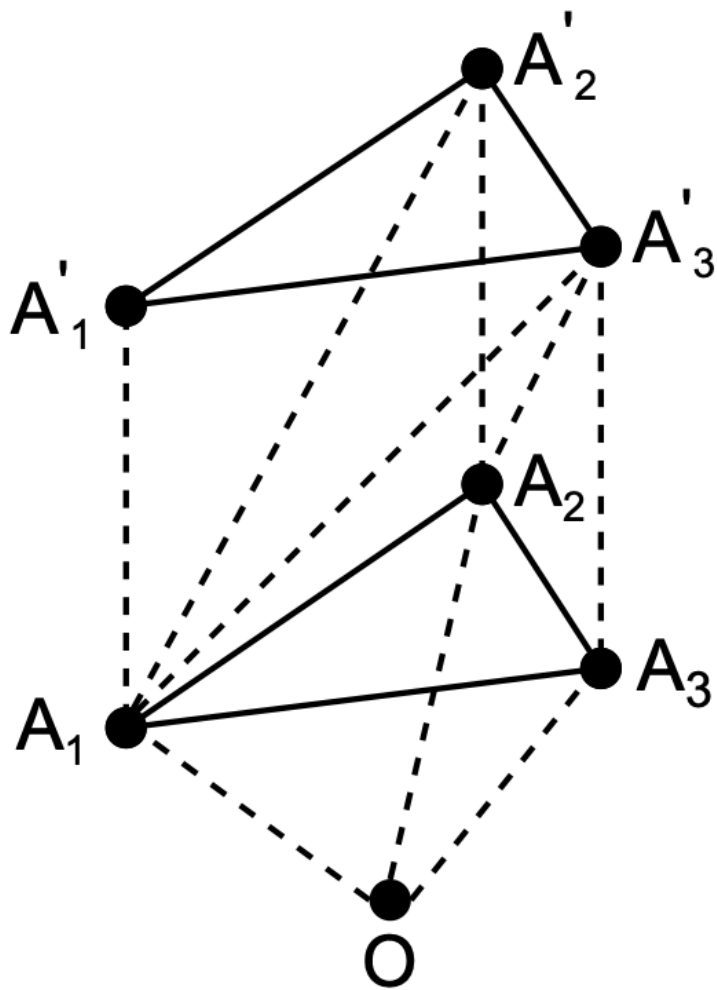
Scalar function topology divergence



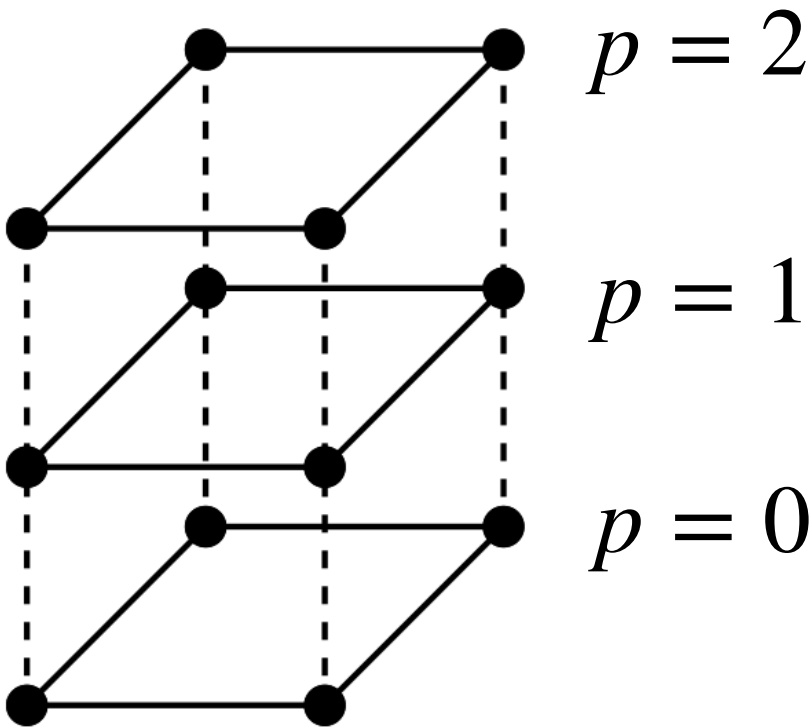
Simplicial complex K



Cubical complex K



$$h(\sigma) = \begin{cases} f(i), & \sigma \in A_i, \\ \min(f(i), g(i)), & \sigma \in A'_i \\ \min(f, g), & \sigma = O. \end{cases}$$



$$h(\sigma, p) = \begin{cases} \min(f, g), & p = 0, \\ \min(f(i), g(i)), & p = 1 \\ f(i), & p = 2. \end{cases}$$

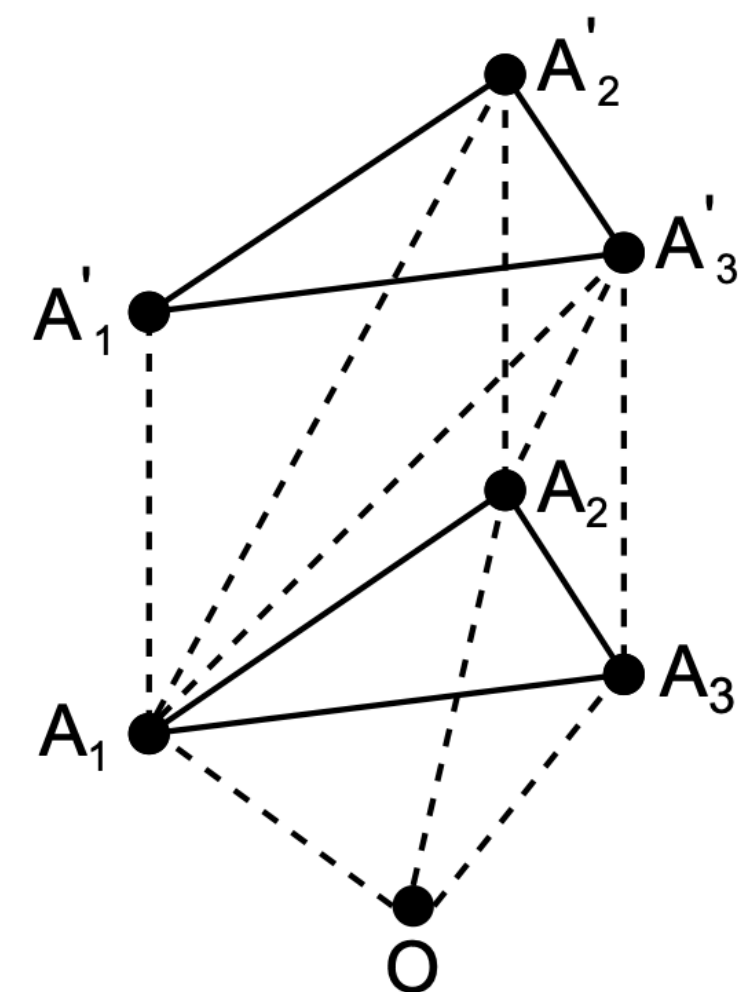
Scalar function topology divergence

Cross-barcode

Cross-barcode

$$CB : (f, g) \rightarrow D(K_h)$$

Each interval in the $CB_k(f, g)$ denotes a range of functions' sublevel sets where a given topological feature exists in the function f but doesn't exist in the function g .



$$h(\sigma) = \begin{cases} f(i), & \sigma \in A_i, \\ \min(f(i), g(i)), & \sigma \in A'_i, \\ \min(f, g), & \sigma = O. \end{cases}$$

Cross-barcode “norm”

$$\text{SFTD}_k^p(f, g) = \sum_i^{|CB_k(f, g)|} \|d_i - b_i\|^p$$